Mission-Enabling Nuclear Electric Propulsion Revisited: an audited, reconciled and optimal-control reanalysis of the SPC 536 study

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Abstract

Nuclear electric propulsion (NEP) removes the power limit that constrains electric propulsion far from the Sun. Our 2025 course paper argued that it is therefore mission enabling, including a 5 t Mars cargo delivered in 200 days by a 100 kWe reactor. An audit of that paper found 44 errors, and this paper rebuilds the analysis. The original calculation was first reproduced and then corrected: the NEP transfer time is 3.00 yr, not 0.85, and the 200-day mission cannot be closed. A trajectory simulation and a SMAD-style subsystem study, which disagreed, were then run on one documented input set together with a new optimal-control solver and a seeded Monte Carlo. The simulation's technology-tier Mars results were found to come from failed trajectory-table nodes. Once these are removed, the two models agree to 5–15 % for Mars (8 t in 450 d needs 95, 64 and 80 kWe in the 2020s, 2030s and 2040s), and for the outer planets they differ only through the choice of capture definition. Burn-coast-burn trajectories are upper bounds, with trip times 3–46 % longer than the optimum at equal propellant. Verdicts are given in two parts. Technical closure never worsens with a better technology state: Mars closes in every state, Jupiter from the 2030s state and Uranus with low margin in the 2030s and fully in the 2040s state. Programmatic readiness then gives the earliest launch: Aug 2039 for Mars (gated by the reactor ground test), 2041 for Jupiter and 2045–2051 for Uranus (gated by converter life tests of 70,000 and 105,000 h). Against current alternatives, NEP is not faster: one Falcon Heavy places 6.8 t in the same Mars orbit in 259 d, Europa Clipper reaches Jupiter in 5.5 yr, and a solar-electric Uranus mission launched in 2035–2040 would arrive about a decade before the NEP orbiter. NEP's gains are mass into a chosen orbit and tens of kilowatts at the destination, which raise the Uranus data rate tenfold, at an estimated flagship cost of $4.1–11.1 billion. Throughout, results marked *computed* are model outputs, *literature* values are cited, and the 2030s and 2040s technology states are *assumed* milestones, not forecasts.

Contents

Nomenclature **2**

1 Introduction **2**

2 Approach **3**
2.1 Propulsion and mass closure . . . 4
2.2 Power sources and thermal sizing . . . 4
2.3 Trajectory models . . . 4
2.4 Launch, uncertainty and verdict framework . . . 5
2.5 Inputs, sources and status . . . 5
2.6 Key assumptions . . . 5

3 Model Validation **6**

4 Results **7**
4.1 Reproducing the original study (Case A) . . . 8
4.2 Corrected analysis of the same missions (Case B) . . . 8
4.3 Reconciling the two models . . . 9
4.4 Feasible redesign: what changes and why . . . 11
4.5 Thermal and radiator design . . . 12
4.6 SMAD subsystem sizing and technology readiness . . . 14
4.7 Operations-level design: power flow, data, control, reliability, operations and cost . . . 16

4.7.1 How the power system works . . . 16
4.7.2 What the mission is for, and how its data get home . . . 17
4.7.3 Reliability and redundancy over a decade . . . 18
4.7.4 Deployment sequence, nuclear safety and mission timeline . . . 19
4.7.5 Cost estimate . . . 19
4.8 Mission results: technical closure and programmatic readiness . . . 25
4.9 Comparison with flown missions and current alternatives . . . 27
4.10 Novel methods: what holds . . . 32
4.11 Sensitivity and "what would need to be true" . . . 33

5 Discussion **34**
5.1 What a reviewer would challenge . . . 35
5.2 Sources of error and model limitations . . . 37

6 Conclusion **37**

Acknowledgments **38**

A Reproducibility Statement **38**

Nomenclature

Propulsion Parameters

F	thrust [N]
$\dot{m}$	propellant mass flow rate [kg/s]
v_e	effective exhaust velocity [m/s]
I_{sp}	specific impulse [s]
g_0	standard gravitational acceleration [m/s^2]
η_t, η_{ppu}	thruster efficiency and power processing unit (PPU) efficiency
P_{thr}, P_{jet}	electric power to the thrusters and jet power [W]
a_0	initial thrust acceleration, F/m_0 [m/s^2]

Power System Parameters

P_e, P_{bus}	electrical power output and bus load [W]
f_{EP}	fraction of electrical power allocated to propulsion
α	power-system specific mass [kg/kWe] (in Eq. (8), the steering angle [rad])
τ, λ	RTG decay time constant [yr] and NEP degradation rate [yr^{-1}]
P_{waste}, Q_{rej}	waste heat to be rejected [W]
A_{rad}, T_r, T_s	radiator area [m^2], radiator temperature and sink temperature [K]
ε, σ, F_{eff}	emissivity, Stefan–Boltzmann constant, effective view factor

Mission Parameters

ΔV	velocity change required for the mission [km/s]
C_3	launch energy, the square of the hyperbolic excess speed [km^2/s^2]
m_0	initial spacecraft mass at Earth escape [kg]
m_p, m_{pl}, m_{bus}, m_{ps}, m_{thr}	propellant, payload, bus, power-system and thruster masses [kg]
c, f_{tank}	unallocated reserve and tank mass fraction
T, t_{burn}, t_{trip}	transfer time, thrusting time and trip time [d or yr]
P_{close}	probability that a mission closes in the Monte Carlo

1 Introduction

Electric propulsion offers an order of magnitude more specific impulse than chemical propulsion, but beyond the inner solar system it is limited by the power that solar arrays or radioisotope generators can supply rather than by thruster physics. Nuclear electric propulsion (NEP) replaces these sources with a fission reactor whose output does not fall with distance from the Sun. The 2025 course paper [1] asked whether NEP is therefore "mission enabling". Its headline claims were that NEP transfers are about twenty times faster than solar-electric ones, that a 100 kWe NEP system makes a 5 t, 200-day Mars cargo mission possible, and that NEP has the smallest specific mass of the power sources compared. A line-by-line audit of that paper (44 findings: 7 critical, 18 major, 13 minor, 6 typographical; `audit/audit_report.md`) found that the qualitative argument partly survives, since a decaying RTG cannot power a fast mission and NEP power

does not fall with distance. The quantitative conclusions, however, do not survive. The errors that change a conclusion are:

- **Table 1 is inconsistent with the paper's own Eq. (26).** The NEP time printed as 0.85 yr is 3.16 yr from the equation (ratio 0.27), and the solar and RTG columns are all scaled by an unexplained factor of about 0.53. Because time is inversely proportional to power for a fixed thruster, the "20× faster" claim is therefore 10× at best.
- **The constant-mass propellant law** (Eq. 19/26) gives a resistojet propellant load of 207 % of the spacecraft mass, so the "73-day Mars transfer" is physically impossible.
- **The Mars case study (Table 3) mixes alpha conventions** (per kWe versus per kW of thruster power) and omits the thrusters, PPU, tanks and bus. It concludes that "solar is prohibitive" when its own table shows solar needing only 17 % more power and 18 % more initial mass. The 200-day transit is also shorter than the 259-day Hohmann flight time, and the empirical $\Delta V(T)$ law (Eq. 27) falls below the Hohmann minimum for $T > 220$ d.
- **The RTG decay constant** (75 yr) is not that of the MMRTG, which is 40.1 yr (110 We falling to 72 We in 17 yr [2]). With the paper's own decay law, a ten-RTG craft can never deliver the energy it needs (ratio 0.35).
- **The abstract's "smallest specific mass"** is contradicted by the paper's own solar value, and no outer-planet mission is ever computed, although the argument for NEP rests on distance.

There therefore remains a gap between what the original study claims and what its own equations and inputs support, and no consistent quantification of where NEP is in fact enabling. The goal of this paper is to address this gap. This study rebuilds the analysis in four steps and keeps every number traceable to a model output or a source.

The rest of the paper is structured as follows. Section 2 sets out the equations, inputs and verdict rule, and Section 3 validates the models. In the Results (Section 4), the original calculation is first reproduced exactly and then corrected (Sections 4.1–4.2). Two independently built models, a trajectory-integrating simulation (`sim`) and a SMAD-style subsystem design study (`smad`), which disagreed in places, are then reconciled on a single documented input set, and their trajectory sub-models are checked against a new optimal-control solver (Section 4.3). Next, a feasible redesign and a thermal and subsystem analysis are presented for three technology eras (Sections 4.4–4.6), followed by the operations-level analyses that SMAD expects (power flow, communications and data, command and data handling, attitude control, reliability, operations and cost; Section 4.7), the two-part mission verdicts with earliest launch years (Section 4.8) and a comparison with flown missions and current alternatives (Section 4.9). The design ideas that survive a paired Monte Carlo test and a literature check are then listed with their gains and precedents (Section 4.10), together with the sensitivities (Section 4.11). The Discussion (Section 5) covers the limitations and the strongest objections a reviewer would raise, and the paper concludes with an assessment of where NEP is and is not enabling (Section 6). Throughout, we mark which results are computed by the models, which are literature values and which depend on speculative technology assumptions: *computed* (model output), *literature* (a value read in a source), *assumed* (a stated assumption with a range).

2 Approach

The goal of this study is to determine, with consistent inputs, when replacing solar arrays or radioisotope thermoelectric generators (RTGs) with a fission power system makes an electric-propulsion mission possible. The analysis links

1. the power source (solar, RTG or NEP) and its thermal sizing,
2. the electric thrusters and the mass closure of the vehicle,
3. a heliocentric trajectory model with a planetocentric capture, and
4. the launch capacity, an uncertainty model and a verdict rule,

to compute power, mass, trip time and probability of closure for each mission and technology era. The equations and inputs used for each part are given below.

2.1 Propulsion and mass closure

Thrust and mass flow follow from the jet-power relation $P_{jet} = \frac{1}{2} F v_e = \eta_t P_{thr}$ with $v_e = g_0 I_{sp}$ [3]:

$$F = \frac{2\,\eta_t\,\eta_{ppu}\,f_{EP}\,P_e}{g_0 I_{sp}}, \qquad \dot{m} = \frac{F}{g_0 I_{sp}}, \tag{1}$$

$$f_{EP} = 1 - \frac{P_{bus}}{P_e}, \qquad m_p = m_0 \left(1 - e^{-\Delta V / v_e}\right), \tag{2}$$

$$t_{burn} = \frac{m_p\, v_e^2}{2\,\eta_t\, P_{thr}} \quad (P_{thr}, I_{sp} \text{ constant}). \tag{3}$$

where F is the thrust, $\dot{m}$ the mass flow, P_e the electrical power, m_p the propellant mass and t_{burn} the thrusting time. The original paper's Eq. (26) is the same expression with $m_p \to m_0 \Delta V / v_e$ (constant mass), which overstates the propellant by up to a factor of two. We keep the PPU and thruster efficiencies separate (η_{ppu} is 0.90 for NSTAR [4] and 0.93 in the SMAD study). The generic thruster efficiency is $\eta_t(I_{sp}) = \eta_{max}/(1 + (I_{sc}/I_{sp})^2)$ with $\eta_{max} = 0.763$ and $I_{sc} = 1195$ s; it was fitted to six verified operating points and is used only between 1800 and 4300 s. The mass of the vehicle at Earth escape then closes as

$$m_0 = m_{pl} + (1 + c)\,(m_{bus} + m_{ps}(P_e) + m_{thr}) + (1 + f_{tank})\,m_p, \tag{4}$$

where c is the unallocated reserve (0.10, the SMAD system margin), $f_{tank} = 0.10$ [5], $m_{thr} = 3.57$ kg/kW of thruster rating plus a PPU of 5 kg/kW, and the power-system mass m_{ps} is read from the tier tables of the SMAD study (Section 4.6). The thruster count is taken as the larger of $\lceil P_{thr}/P_{unit} \rceil$ and the number that keeps the hours and xenon throughput per unit below 80 % of the demonstrated values (NEXT: 51,184 h, 918 kg [6]; NSTAR: 30,352 h, 235 kg [7]).

2.2 Power sources and thermal sizing

Solar array power is modelled with an inverse-square law in distance and an exponential degradation in time, RTG power with an exponential decay, and NEP power with a slow exponential degradation from fuel burnup and component ageing. The waste heat of the fission system sets the radiator area:

$$P_{solar}(r,t) = P_{1\,AU} \left(\tfrac{1\,\mathrm{AU}}{r}\right)^2 e^{-\lambda_s t}, \qquad P_{RTG}(t) = P_0\, e^{-t/\tau}, \quad \tau = \frac{17\ \mathrm{yr}}{\ln(110/72)} = 40.1\ \mathrm{yr}, \tag{5}$$

$$P_{NEP}(t) = P_0\, e^{-\lambda t}, \quad \lambda = -\ln(0.95)/15\ \mathrm{yr} = 3.42 \times 10^{-3}\ \mathrm{yr}^{-1}, \qquad P_{waste} = P_e \left(\tfrac{1}{\eta_{sys}} - 1\right), \quad A_{rad} = \frac{P_{waste}}{\varepsilon \sigma F_{eff}\,(T_{rad}^4 - T_s^4}$$

(6)

where τ is the RTG decay constant, λ the NEP degradation rate and T_{rad} and T_{sink} the radiator and sink temperatures. It follows that an RTG can deliver at most $P_0 \tau$ of energy over its life. The radiator view factor $F_{eff} = 1.50$ is calibrated to the 26.9 m^2 of the El-Genk–Tournier design [8]; in the SMAD study the area also includes a 10 % margin and a fin efficiency (Section 4.5).

2.3 Trajectory models

All trajectory models treat planar heliocentric two-body motion with thrust from Eq. (1). Departure is at Earth escape, with hyperbolic excess speed $\sqrt{C_3}$ along Earth's velocity. Arrival is defined as rendezvous with the *orbit* of the target planet (circular, coplanar, semi-major axis from [9]) with free phase, so launch-window waiting is not costed. It is followed by a tangential capture spiral from rest to a circular orbit of radius r_c (20 planetary radii for Mars, 30 for the giants):

$$\Delta V_{cap} = \sqrt{\mu_p / r_c}, \qquad t_{cap} = \frac{m_{p,cap}\, v_e^2}{2 \eta_t P_{thr}}. \tag{7}$$

Three heliocentric models are compared in Section 4.3.

(a) Burn-coast-burn family (sim). This model flies a velocity-aligned burn, a Kepler coast and a second velocity-aligned burn, and is solved by a one-dimensional shooting in the junction semi-major axis. The resulting trajectory is feasible but not optimal.

(b) The SMAD $\Delta V(T)$ law. The SMAD study uses $\Delta V_{EP}(T, C_3) = \sqrt{\Delta V_\infty^2 + \beta^2/T^2} + \sqrt{C_{3,ref}} - \sqrt{C_3}$, with $\beta^2 = (\Delta V_{ref}^2 - \Delta V_\infty^2) T_{ref}^2$ calibrated at one point per case, and $t_{trip} = t_{burn}/\text{duty}$. The law includes the capture, but it is a heuristic.

(c) Optimal control (new). The new solver uses the state (r, θ, v_r, v_t, m) in canonical units (1 AU, $\sqrt{\text{AU}^3/GM_\odot}$), with the steering angle α from the local horizontal and the throttle $u \in [0, 1]$ as controls:

$$\dot{r} = v_r, \qquad \dot{\theta} = v_t/r, \qquad \dot{v}_r = \frac{v_t^2}{r} - \frac{1}{r^2} + \frac{uF}{m}\sin\alpha, \quad \dot{v}_t = -\frac{v_r v_t}{r} + \frac{uF}{m}\cos\alpha, \quad \dot{m} = - \tag{8}$$

$$F = \frac{2\eta_t P_{thr0}}{v_e}\left(\tfrac{1\,\text{AU}}{r}\right)^n e^{-\lambda t}, \qquad n = 0 \text{ (fission)},\ n = 2 \text{ (solar)}.$$

The boundary conditions are $r(0) = 1$, $v_t(0) = v_c + \sqrt{C_3}$, $v_r(0) = 0$ and $r(T) = r_t$, $v_r(T) = 0$, $v_t(T) = \sqrt{1/r_t}$. Power-limited thrust at constant I_{sp} is bang-bang, so the *minimum-time* problem uses $u \equiv 1$ and the *minimum-propellant* problem at fixed $T \geq T_{min}$ lets u drop to zero on coast arcs. The controls are piecewise linear in normalised time (14–20 steering nodes, 16–24 throttle nodes) and are integrated with fixed-step RK4 (400–500 steps, numba). The nonlinear program is solved with SLSQP from 45 steering starts, followed by a refinement with 20 nodes. A solution found this way is a *feasible* trajectory and therefore an *upper bound* on the true minimum time and propellant; it is not a proof of global optimality (Section 3). For mass closure, the solutions are tabulated against the initial acceleration $a_0 = F/m_0$, the exhaust velocity and the time ratio $s = T/T_{min}$ (9×3×6 nodes per target and C_3) and then interpolated. This is exact for a distance-independent source, because the heliocentric motion then depends only on a_0 and v_e.

2.4 Launch, uncertainty and verdict framework

Launch capacity is modelled as $m(C_3) = m_{LEO} \exp(-\Delta V(C_3)/v_{e,eff})$, with $\Delta V(C_3) = \sqrt{C_3 + 2\mu/r} - \sqrt{\mu/r}$ at $r = 6778$ km. The pair $(m_{LEO}, v_{e,eff})$ is (63.8 t, 2.39 km/s) for an expendable Falcon Heavy (fitted to 10.9 t at $C_3 \approx 24$ [10]), (105 t, 3.2 km/s) for SLS Block 1B (availability uncertain) and (100 t, 4.0 km/s) for a refuelled Starship (a pure assumption). A 10 % vehicle margin is included in the capacity, and a further 1.15 design reserve is applied. Closure probabilities are computed with a Latin hypercube Monte Carlo ($N = 2000$, seed 20251103) over the power-system mass multiplier, I_{sp} (±5 %), thruster efficiency (0.90–1.02), PPU efficiency (0.84–0.94), thruster life (×0.5–1.5; 0.25–2 for the 2040s), launch capacity (×0.85–1.05), bus mass (×0.9–1.25) and degradation (×1–3), with the hardware fixed and the trajectory re-planned. A mission *closes* in a sample when the propellant fits, the trip time is within the limit, the thruster life and throughput hold, the stack fits the vehicle and the radiator is within the largest studied (2,500 m^2 [11]). The verdict is given in two separate parts (Section 4.8). The *technical closure* uses only this physical closure: a technology state *closes* a mission if $P_{close} \geq 0.8$, closes *with low margin* if $P_{close} \geq 0.5$, and otherwise does not close. The *programmatic readiness* of the same state is the year in which its critical items could reach TRL 6, its converter could have run for the mission's full-power life, a launch vehicle for the stack could exist and the flight unit could be integrated. An earlier version of this study merged the two through a single rule (POSSIBLE required TRL ≥ 6), which ranked the lighter 2040s designs below the 2030s ones because the newer technology state carries TRL-5 items; the two-part form removes that inversion. The thresholds 0.8 and 0.5 are our judgement, not a standard.

2.5 Inputs, sources and status

Table 1 lists the principal inputs. Every number in the models is read through a parameter store that records its source and a status: VERIFIED means a document was opened and the value seen, DERIVED means computed from verified values, and UNVERIFIED means secondary, search-snippet only or an assumption. Of the 148 keys read by the simulation, the UNVERIFIED ones are marked in `sim/results/parameters_used.json`, and the tier assumptions of the SMAD study are kept in its assumption register (`smad/results/assumption_register.csv`).

2.6 Key assumptions

Finally, we summarise the key modelling assumptions, each of which is discussed again in Section 5:

- Each thruster operates at a fixed I_{sp} and efficiency; no throttle curves are modelled.

Table 1: Selected inputs read from `data/params.json` and `sim/extra_params.json`; the full list with status for every key is in `sim/results/parameters_used.json`. Source ids refer to `data/sources.md`.

Input	Value	Source id	Status
NSTAR Isp, total efficiency	3270 s / 0.65	[5]	VERIFIED
NSTAR PPU efficiency	0.9	[5]	VERIFIED
NSTAR demonstrated life / throughput	30352 h / 235 kg	[6]	VERIFIED
NEXT Isp, total efficiency	4170 s / 0.706	[7]	VERIFIED
NEXT demonstrated life / throughput	51184 h / 918 kg	[8]	VERIFIED
MMRTG power after 17 yr (BOL 110 We)	72 We	[16]	VERIFIED
MMRTG specific mass	409 kg/kWe	[16] derived	DERIVED
Dawn array power at 1 AU	10.3 kW	[30]	VERIFIED
solar array degradation	0.0075 /yr	[1] (terrestrial basis)	UNVERIFIED
solar array specific mass	[25, 40] kg/kW	[37]	VERIFIED
fission power decay rate	0.00342 /yr	[2] design goal	DERIVED
housekeeping bus load (sim)	5 kWe	assumption	UNVERIFIED
capture orbit, planets / giants	20 R_eq / 30 R_eq	assumption	UNVERIFIED
largest radiator studied	2500 m^2	[21]	VERIFIED
propellant tank mass fraction	0.1 tank mass / propellant mass	SMAD Table 10-12	UNVERIFIED
launch-vehicle margin	0.1	assumption	UNVERIFIED

- Planets are on circular, coplanar orbits with free phase; launch-window waiting, inclination and Earth escape from low orbit are not costed.
- Optimal-control solutions are feasible trajectories and are treated as upper bounds on the true minimum time and propellant.
- Power-system masses are taken from the conceptual tier tables of the SMAD study; the launch curves for SLS Block 1B and a refuelled Starship are partly or wholly assumed.
- The 2030s and 2040s technology tiers are *assumed* states conditional on stated milestones, not forecasts, and the verdict thresholds are our judgement.
- The development durations that set the earliest launch years (reactor ground test, converter life test, integration) are taken from the SMAD study's plan and are assumptions (A); their sensitivity is given in Table 18.

3 Model Validation

Before the models are used for any mission result, each is checked against analytic solutions, flight data or published designs. This section summarises those checks and states what they do not cover.

Simulation (sim). The simulation passes 59 of 59 automated checks (Table 2). The ballistic energy and angular-momentum drift is of order 10^{-11} over five orbits, the integrated Hohmann ΔV is within 10^{-4} of the tabulated values, the Tsiolkovsky equation holds to machine precision, and the thrust-power law is within 0.03 % of the measured NSTAR thrust. The power-system presets reproduce the El-Genk 3390 kg / 26.9 m^2 design and the Mason 4115 kg Brayton design. Finally, a Dawn-like benchmark (a planar spiral from 1 AU to Ceres with the NSTAR throttle table) returns 385 kg of xenon against the 416 kg consumed by Dawn [12], a ΔV of 11.6 against 11.5 km/s, and 39,700 against 51,250 thrust hours. The 22 % deficit in hours comes from the assumed throttle table and an unverified housekeeping load, and the match in ΔV is partly fortuitous, because Dawn's Mars flyby and the omissions of the planar model offset each other.

Limits of that validation. The check suite covers the trajectory table (V6, 2.6 % interpolation error) only at the nodes that the integrator solved. It did not test whether every node was solved. Section 4.3 shows that, for two of the tables used in the technology-tier study, 35 of 63 and 7 of 63 nodes had failed and had been filled with neighbouring values, an error that the suite could not detect.

SMAD study. The power-system, shield, thermal and mass models of the SMAD study are validated against the JIMO Brayton design (the shield model is conservative by about a factor of 1.9 on mass), against the NASA 20 kWe Uranus mass-equipment list [10] and against Mason's 20/100/200 kWe specific-mass curve [13], with differences of −28 %, +6 % and +25 % respectively. Above about 300 kWe the model flattens at 40 kg/kWe, whereas the literature keeps falling (13 kg/kWe at 1.9 MWe [14]), so it must not be used in that range.

Table 2: Simulation validation: largest relative error by family (59 checks, all within tolerance).

Family	Check	Max rel. error
V1 tangential	integrated tangential-spiral ΔV vs $\lvert v_1 - v_2 \rvert$	0.0029
V2 angular-momentum	angular momentum, 5 orbits	6.3e-12
V2 energy	energy, 5 orbits	1.7e-11
V2 position	position, 5 periods	8e-10
V3 Hohmann	Hohmann ΔV vs params	0.00011
V3 integrated	integrated Hohmann apoapsis	3.9e-12
V4 integrated	Tsiolkovsky, integrated	3.9e-15
V4 propellant	propellant fraction	5e-16
V5 RK4	RK4 vs analytic	8e-15
V5 spiral	analytic radius-from-ΔV model	0.018
V6 table	trajectory-table interpolation	0.026
V7 high-thrust	burn-coast-burn $\rightarrow$ Hohmann	0.0058
V7 low-thrust	burn-coast-burn $\rightarrow$ spiral	0.00033
V8 AMTEC	AMTEC preset vs El-Genk	9.5e-05
V8 Brayton	Brayton preset vs Mason	2.2e-16
V8 MMRTG	MMRTG 72 We at 17 yr	0
V8 scaling	power-system scaling law	0.092
V8 solar	solar array law	1.9e-16
V8 thermal	radiator equation	1.3e-16
V8 thrust-power	thrust-power law vs NSTAR 92 mN	0.00026
V9 Dawn:	Dawn-like benchmark (thrust hours)	0.22

Table 3: Checks of the optimal-control solver (`reconcile/oc_validate.py`).

Optimal-control solver check	Value
Coast, eccentric orbit, 5 periods: energy drift	2.4e-12
Hohmann coast, apoapsis radius error	2.4e-14
Low-thrust limit, a_0 = 3e-05 m/s^2: ΔV / Edelbaum -1	+0.025
Mars, N=300, K=10: T_{min} = 265.6 d, DOP853 terminal error	1.1e-07
Mars, N=400, K=14: T_{min} = 265.2 d, DOP853 terminal error	1.2e-06
Mars, N=500, K=20: T_{min} = 265.1 d, DOP853 terminal error	5.2e-06
Mars, N=700, K=28: T_{min} = 264.9 d, DOP853 terminal error	8.3e-07
Jupiter, N=300, K=10: T_{min} = 768.6 d, DOP853 terminal error	3.9e-08
Jupiter, N=400, K=14: T_{min} = 768.5 d, DOP853 terminal error	1.9e-06
Jupiter, N=500, K=20: T_{min} = 768.2 d, DOP853 terminal error	7.3e-07
Jupiter, N=700, K=28: T_{min} = 768.2 d, DOP853 terminal error	1.3e-06
Propellant fraction at 2 T_{min} vs impulsive Hohmann (0.1601)	0.1606

Optimal-control solver (new). Table 3 and the checks below test the solver itself. (i) With zero thrust, the integrator holds a circular orbit for five periods to $2e-12$ in radius and reproduces a Hohmann half-ellipse to 2e-14 in apoapsis radius (flight time 259 d, the analytic 259-d value). (ii) The optimal controls, re-integrated with an independent adaptive integrator (DOP853, relative tolerance 10^{-11}), miss the target orbit by at most 5e-06 in canonical units; the fixed-step RK4 shooting is therefore not an artefact. (iii) In the convergence test on the number of steering nodes and steps, the minimum times differ by 0.10 % between 14 and 28 nodes. (iv) In the low-thrust limit ($a_0 \rightarrow 0$), the flown ΔV approaches the Edelbaum value $\lvert v_{c1} - v_{c2} \rvert$ to within 2.5 %. (v) With a time budget of twice the minimum time, the propellant fraction tends to 0.161, against 0.160 for the impulsive Hohmann transfer. (vi) Every optimal solution found is a feasible trajectory, so $T_{opt} \leq T_{bcb}$ must hold wherever the burn-coast-burn solution is valid; this held in all comparisons of Section 4.3. Two things are *not* validated. The first is global optimality: a multi-start search is used, and 12 of 18 starts agree on the same optimum in the test case, but there is no indirect-method certificate. The second is the planetocentric capture, which is analytic. The optimal-control times should therefore be treated as upper bounds on the true minima.

4 Results

This section applies the framework of Section 2 in the order in which the study was carried out. The original calculation is first reproduced (Case A) and corrected (Case B); the two independent models are then reconciled, a feasible redesign and its thermal and subsystem sizing are presented, and the two-part mission verdicts, the comparison with current alternatives, the design ideas that survive testing and the sensitivities follow.

Table 4: Case A: Table 1 of the paper re-computed from its own Eq. (26); last column is the corrected model (rocket equation, PPU 0.90, decaying RTG; "never" = the source cannot deliver the energy).

Row (10 km/s, 20 t, NSTAR)	Paper [yr]	Eq. 26 [yr]	Paper / Eq. 26	Corrected [yr]
Single MMRTG	1900	3588	0.53	never
Cassini (3 RTGs)	240	446	0.54	never
Scaled (10 RTGs)	190	359	0.53	never
Solar (Dawn) @1 AU	17	32	0.54	33.79
Solar @5 AU	420	789	0.53	never
NEP 100 kWe	0.85	3.16	0.27	3.00

Table 5: Case A reproduces the paper's Table 3 to its printed precision (rows 1–2). Case B applies the audit corrections to the same mission definition (AMTEC power-system model, PPU, thrusters, tanks, 1 t bus): the required electric power is 250 kWe, 27% payload fraction, and the 200-d definition is shorter than the 259-d Hohmann time.

Mars cargo, 5 t, 200 d	P paper [kW]	P recomputed [kW]	m_{ps} paper [kg]	m_{ps} rec. [kg]	m_0 paper [t]	m_0 rec. [t]
NEP	58	57.7	1904	1904	9.5	9.50
Solar	68	68.1	3146	3145	11.2	11.20
NEP, corrected (Case B)	-	250	-	6008	-	18.5
Solar, corrected (Case B)	-	none	-	-	-	-

4.1 Reproducing the original study (Case A)

We first re-implement the paper exactly as written (the Eq. 26 constant-mass law, no PPU, the Eq. 27 empirical $\Delta V(T)$ and Eq. 28) with its stated inputs, in order to find where each printed number comes from.

Table 1. Table 4 shows that none of the printed times follows from Eq. (26). Five rows are low by a constant factor of 0.53–0.54, and the NEP row by 0.27. Corrected with the rocket equation and a PPU efficiency of 0.90, a 100 kWe NEP craft needs 3.00 yr, not 0.85 yr, and the RTG rows have no solution at all, because the source cannot deliver the energy.

Table 2. The NEP column reproduces exactly from $t = m_0 \Delta V g_0 I_{sp}/(2\eta P)$ (692 d for NSTAR at 50 kW, 422 d for the SPT-100, and so on). The solar column reproduces only if the mean solar power is 3.16 kW rather than the stated 2.2 kW, and the RTG columns cannot be produced from any single power. With the constant-mass law, the resistojet case needs 41,480 kg of propellant for a 20 t craft and the arcjet case 20,394 kg (102 %), which no spacecraft can carry.

Table 3 and Fig. 8. The arithmetic of the Mars cargo table is correct given its inputs (Table 5); the problem lies in the inputs themselves. The paper's own Eqs. (27)–(28) place the infinite-power asymptotes at 112 d (NEP) and 130 d (solar), not at the 150 and 250 d of its Fig. 8, and Eq. (27) returns 2.5 km/s at 500 d, which is below the 5.59 km/s heliocentric Hohmann minimum. Figure 1 redraws the paper's Figs. 4, 6, 7 and 8 with these corrections.

4.2 Corrected analysis of the same missions (Case B)

Case B keeps the paper's mission definitions and applies only the audit corrections listed in Table 6; no redesign is done at this stage.

Results. With these corrections, the NEP 100 kWe craft of Table 1 needs 3.00 yr (Table 4). A solar craft of the paper's mass needs 33.8 yr at 1 AU, and at 5 AU or on any RTG the mission never completes. The ratio of solar to NEP time is therefore about 11, not 20. The advantage of NEP is real, but it comes from distance rather than from the nominal factor. Figure 2 shows that, at equal power-system mass, a solar array out-produces a fission system of the same mass out to the crossover distance marked in the figure for the 2020s system, and to about 1.1–1.4 AU for the 2030s and 2040s systems. An RTG system of the same mass is 3 to 15 times weaker than the fission system at every distance.

The Mars cargo case with the paper's definition. With every correction applied and the paper's inputs kept (5 t, 200 d, a 2000 s Hall thruster, $\eta = 0.5$ and a ΔV of 6.25 km/s from Eq. 27), the mission closes only at 250 kWe, 18.5 t initial mass and a 27 % payload fraction. The reference 100 kWe system

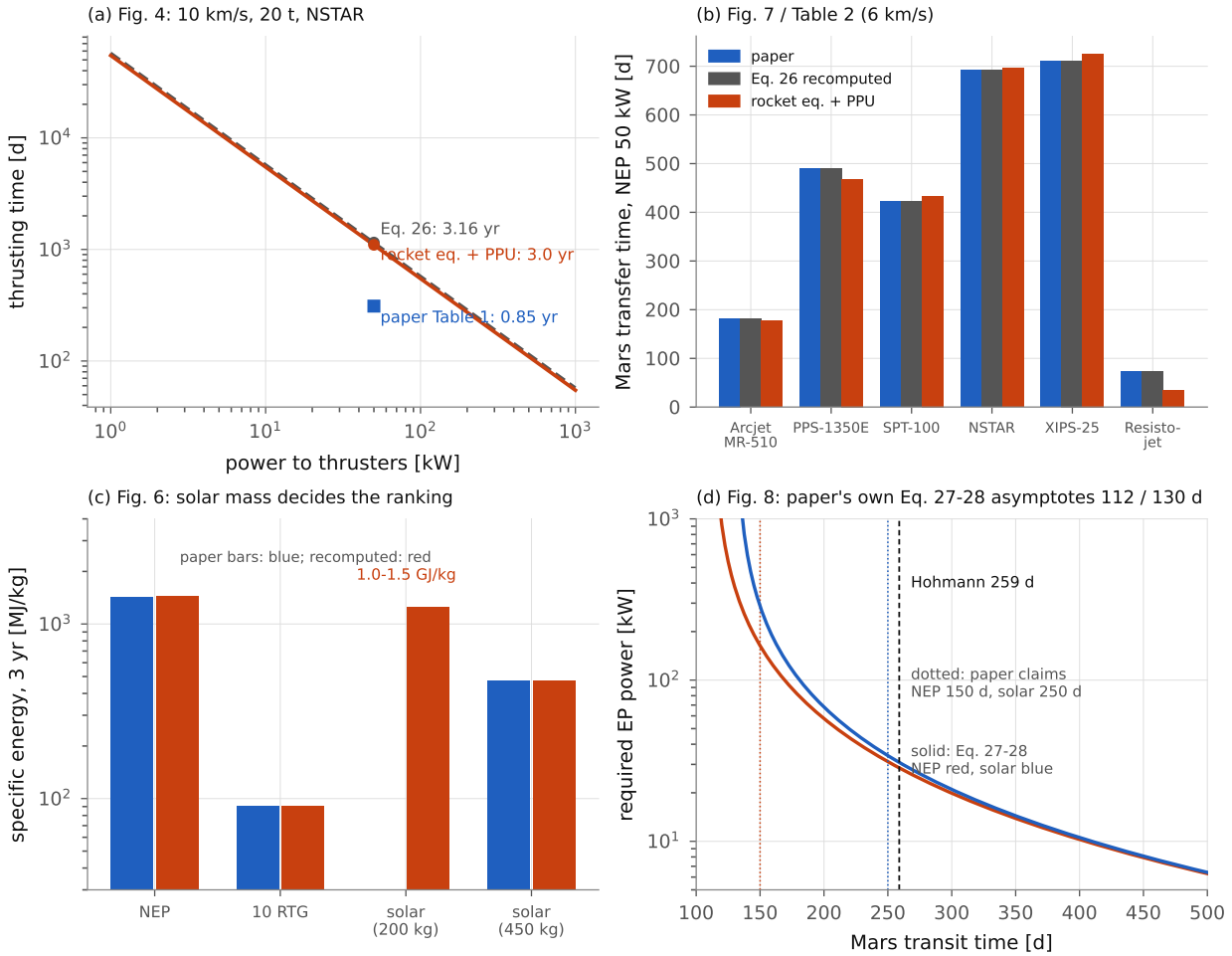


Figure 1: Corrected versions of the original Figs. 4, 6, 7 and 8. (a) NEP time versus thruster power: Eq. 26 and the rocket-equation form agree at the 5 % level for this case, but the printed Table 1 point (blue) is 3.7× too low. (b) NEP column of Table 2: the paper's values reproduce Eq. 26, but the resistojet and arcjet bars are not flyable. (c) Specific energy: the ranking of NEP and solar depends entirely on which solar mass is used. (d) Required electric power versus transit time from the paper's Eqs. 27–28: the asymptotes are at 112 and 130 d, not 150 and 250 d, and every point left of the dashed line is below the Hohmann flight time. None of the four original figures survives the corrections unchanged.

cannot do it (closure probability 0 in 2000 Latin-hypercube samples; required power 236–369 kWe, 5th–95th percentile), and solar-electric propulsion does not close at all. Two further problems are independent of mass: 200 d is shorter than the 259-d Hohmann time, and the ΔV itself is not supported by any source. Section 4.3 uses the optimal-control solver to test whether 200 d is reachable at all.

4.3 Reconciling the two models

The simulation (`sim`) and the SMAD-style design study (`smad`) were built independently, and they disagreed in four places. (1) The simulation found that the 8 t, 450-d Mars cargo mission closes in every technology tier, while SMAD rated it not possible in the 2020s. (2) The simulation's trip times come from a tangential burn-coast-burn family, whereas SMAD uses a $\Delta V(T)$ law calibrated at one point. (3) Launch capacity versus C_3 and the margins were handled differently. (4) The 2030s and 2040s tiers of the simulation were parameterised placeholders. In this subsection we build one common input set, run both trajectory models on it together with the new optimal-control solver, and explain every remaining difference.

Common inputs. The missions are a Mars cargo of 8 t in 450 d ($C_3 = 8.2$ km^2/s^2), a Jupiter orbiter of 1.5 t in 6 yr ($C_3 = 40$) and a Uranus orbiter of 1 t in 10 yr ($C_3 = 40$). The planets are circular and coplanar, and arrival is defined as rendezvous with the planet's orbit followed by a capture spiral to 20 (Mars) or 30 (giants) planetary radii. The ΔV contingency is 5 %, the trip time is designed to 95 % of the limit, and the thrust duty is 0.88, 0.80 and 0.70. The thrusters, the PPU efficiency of 0.93, the bus load, the bus mass without the SMAD system margin, the 10 % reserve and the 10 % tanks are as in Section 2. The launch

Table 6: Principal Corrections from Case A to Case B (full list: `sim/changes_log.md`).

Item	Paper	Corrected	Source
Propellant law	constant mass, $m_0 \Delta V / v_e$	$m_0(1 - e^{-\Delta V / v_e})$	audit F5
PPU	none	$\eta_{ppu} = 0.90$	NSTAR [4]
MR-510 arcjet η	0.45 (inferred)	0.32 from the data sheet midpoint	audit F10
NSTAR I_{sp}, η	3150 s, 0.62	3270 s, 0.65	[4]
RTG decay τ	75 yr	40.1 yr (110→72 We in 17 yr)	[2]
RTG energy	finite times printed	"never" when $E > P_0 \tau$	integral of Eq. 10
NEP mass	33 kg/kWe × 100 kWe lumped	AMTEC split model 3390 kg at 100.8 kWe	[8]
Radiator	27 m^2, T, ε unstated, 12 kg/m^2	T = 650 K, ε = 0.85, 7.67 kg/m^2	[8]
Solar array	200 kg (Table 1) / 450 kg (Figs. 5–6)	25 kg/kW	lecture notes (UNVERIFIED value)
Solar power, Table 2	undeclared 3.16 kW	10 kW at 1 AU, $1/r^2$ along the path	[12]
Mars cargo	α per kW thruster, no PPU/tanks/bus	AMTEC model + PPU + thrusters + 10 % tanks + 1 t bus	audit F4

vehicles are SMAD's three-vehicle catalogue (identical in both packages: 13,011, 31,181 and 37,069 kg at $C_3 = 8.2$ for Falcon Heavy, SLS Block 1B and Starship) with the 1.15 reserve, and the power-system masses come from the tables of the three SMAD eras. The closure engine is the simulation's mass-closure function, and the three trajectory models are plugged into it through one interface.

The common closure reproduces SMAD. With the SMAD $\Delta V(T)$ law as the trajectory model, the common closure returns 57 kWe / 16.2 t for Mars in the 2030s against SMAD's own 60 kWe / 17.2 t, 17 / 5.7 against 18 / 6.1 for Jupiter in the 2030s, and 81 / 23.2 against 96 / 28.0 for Mars in the 2020s (Table 8). The method is therefore a faithful stand-in for the SMAD model to within 5–15 %, and any remaining difference below is due to the trajectory model or to the definitions, not to the closure code.

Finding 1: the simulation's Mars result for the technology tiers is an artefact of failed table nodes. The simulation tabulates burn-coast-burn outcomes against initial acceleration and exhaust velocity, and it fills the nodes where its solver failed with the nearest solved node. For Mars at $C_3 = 8.2$, 35 of the 63 nodes failed, and the whole range $a_0 < 6.3 \times 10^{-4}$ m/s^2 is filled. All NEP designs of interest ($a_0 \approx 10^{-4}$) lie inside this range, where the table returns the impulsive Hohmann values (about 300 d, propellant fraction 0.07–0.08) independently of thrust. The simulation's Mars rows for all tiers (trip 396–402 d, "closes in every tier", P(close) 0.55–0.68) are therefore not results of its trajectory model (errata E-S0). When these nodes are treated as "no solution", the burn-coast-burn family has no Mars solution at $C_3 = 8.2$ for these accelerations, because the family cannot represent a low-thrust transfer that starts with excess speed. The optimal-control solver has no such failure. It finds a 428-d trajectory with 64 kWe in the 2030s and 95 kWe in the 2020s, in agreement with SMAD (60, 96 kWe).

Finding 2: burn-coast-burn is not far from optimal in propellant but is slow. Table 7 and Fig. 3 compare the two families for the same vehicle. At equal trip time, the optimal propellant fraction equals the burn-coast-burn value to within -0.9–0.7 %, because both reach the impulsive plateau. The minimum time of the optimal solution, however, is 3–46 % shorter (median 30 %), and the gap grows with initial acceleration: a burn-coast-burn transfer cannot go below roughly the Hohmann time, whereas continuous steering can. The family is therefore a safe upper bound for the trip time with no more than a few percent propellant penalty, and the simulation's description of it as an upper bound is correct. Figure 5 shows the same effect at mass closure, as trip time against electric power.

Finding 3: the paper's 200-day Mars mission is not reachable. With the paper's thruster (Hall, 2000 s), $C_3 = 0$ and optimal steering, a heliocentric time of 200 d needs an initial acceleration of 6.3e-04 m/s^2 and a propellant fraction above one half ($\Delta V \approx 15$ km/s, not the 6.25 km/s of Eq. 27). With the paper's mass convention of 33 kg per kW of thruster power, the power-system mass alone at that acceleration is $0.41\, m_0$, and the mass equation has no positive solution at any power (none). The simulation's statement that this mission "closes at 250 kWe" (Case B) follows from the unsupported Eq. 27 and is withdrawn.

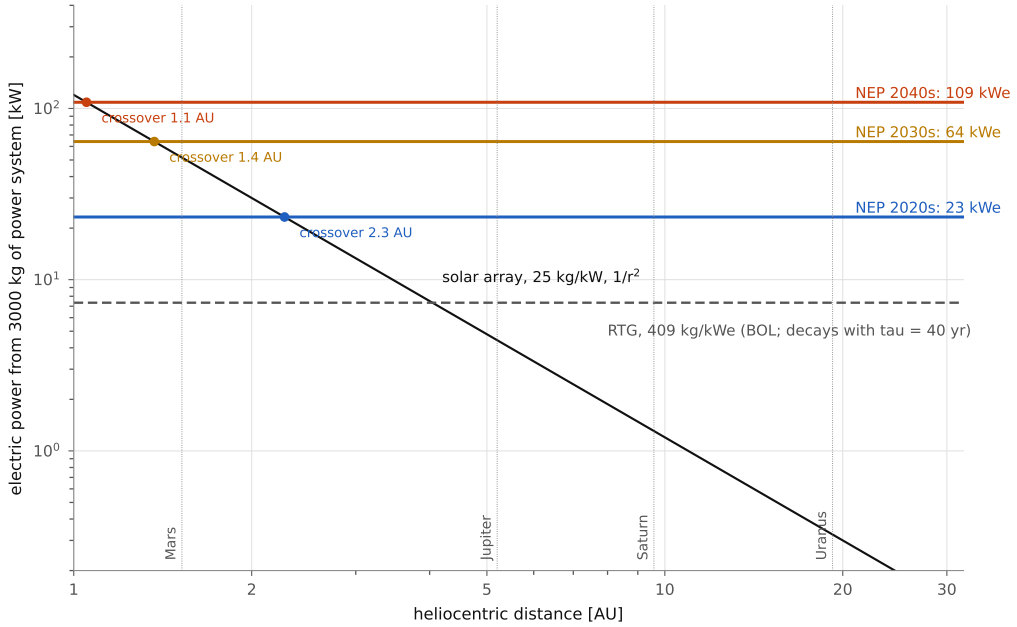


Figure 2: Electric power available from 3000 kg of power system versus heliocentric distance. Solar: 25 kg/kW, $1/r^2$. RTG: 409 kg/kWe at beginning of life. NEP: tier power-system tables of the SMAD study. Dots: crossover with the solar array. Beyond the crossover, the fission systems deliver the most power for a given mass.

Finding 4: the Jupiter and Uranus disagreement is a definition. SMAD's reference ΔV for Jupiter (10 km/s at $C_3 = 30$) contains about 1 km/s of capture, whereas the common definition (rendezvous plus a spiral to 30 planetary radii) needs 7.7 km/s for the spiral alone. With the common definition, the Jupiter 2030s design needs 45 kWe and 9.4 t, against SMAD's 18 kWe and 6.1 t; with a 1 km/s capture, the same closure gives 29 kWe (Table 24). The burn-coast-burn family gives 52 kWe / 9.6 t, within about 15 % of the optimal-control value. This difference favours SMAD's verdicts for the outer planets and is the largest unresolved modelling choice of the study (errata E-M2).

Finding 5: bookkeeping differences in the simulation. The simulation's technology-tier bus mass includes SMAD's 10 % system margin and then applies a second 10 % reserve, while it omits the integration allowance through a stale ledger key; as a result, the bus is 4329 kg against 3278 kg for the 2020s Mars design (errata E-S1, E-S2). The closure-probability conventions also differ (fixed hardware versus re-optimised in each sample), which is why the same Mars design had 0.55 in one study and 0.27 in the other.

Which model is more defensible? For trajectories, the optimal-control model is the more defensible: it gives a feasible solution with no table failures, and it bounds both other models. For the mass model, SMAD's bottom-up ledger (maturity-based margins, structure and thermal floors) is more complete than the simulation's frozen bus plus reserve, but its shield and conversion models are conceptual (Section 3). The two models cannot be reconciled on the outer-planet capture definition, so both numbers are reported. Figure 4 summarises the power and mass of the minimum-mass designs from all models.

4.4 Feasible redesign: what changes and why

Having shown that the original mission cannot be closed as defined, we now look for the lightest design that does close, first with the simulation alone and then for all three missions on the common inputs.

Case C (simulation, Earth escape at $C_3 = 0$, 5 t payload). Starting from the corrected Case B, a deterministic grid search was run over thruster (NEXT, NSTAR, HERMeS) and electric power (40–2000 kWe) for the minimum initial mass under a margin policy: initial mass ≤90 % of the advisory 20 t limit, trip ≤85 % of 600 d, thruster life and throughput ≤80 % of demonstrated, and a radiator within the largest studied. The search gives a NEXT cluster of 71.9 kWe with 11,797 kg at escape, 1715 kg of xenon and 2748 kg of power system (radiator 19 m^2, 242 kW rejected). It uses nine thrusters at 0.18 of the demonstrated life and flies a 492-d trip (burn 228 d, coast 101 d, burn 118 d, capture 45 d; ΔV 5.63 plus 0.79 km/s). Table 9 lists the values changed and the reason for each. In a Monte Carlo with the hardware

Table 7: Burn-coast-burn (sim) versus optimal control for a constant-power vehicle, I_{sp} 4170 s. "solver failed" = the sim table node was filled with a neighbour's value (errata E-S0).

Target	C_3	a_0 [m/s^2]	T_{bcb} [d]	$T_{opt,min}$ [d]	time saved [%]	ϕ_{bcb}	ϕ_{opt} at T_{bcb}
Mars	0	5e-05	1321	1288	3	0.129	0.130
Mars	0	1e-04	782	711	9	0.129	0.129
Mars	0	2e-04	420	381	9	0.128	0.128
Mars	0	4e-04	338	270	20	0.128	0.128
Mars	0	8e-04	297	194	34	0.128	0.128
Mars	8.2	5e-05	solver failed	962	-	-	-
Mars	8.2	1e-04	solver failed	434	-	-	-
Mars	8.2	2e-04	solver failed	292	-	-	-
Mars	8.2	4e-04	solver failed	224	-	-	-
Mars	8.2	8e-04	277	171	38	0.065	0.065
Jupiter	0	1e-04	2506	2073	17	0.332	0.333
Jupiter	0	2e-04	1759	1216	31	0.328	0.328
Jupiter	0	4e-04	1265	715	43	0.315	0.315
Jupiter	40	5e-05	solver failed	2329	-	-	-
Jupiter	40	1e-04	1556	1162	25	0.200	0.199
Jupiter	40	2e-04	1180	820	30	0.188	0.188
Jupiter	40	4e-04	1083	587	46	0.182	0.184

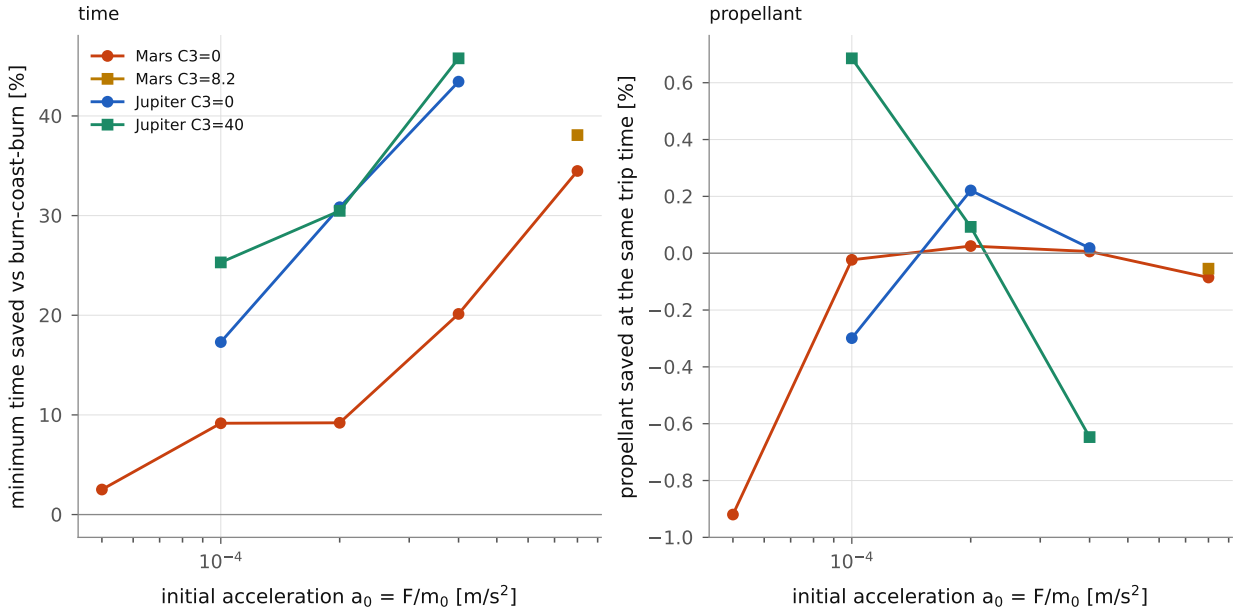


Figure 3: Optimal control versus the burn-coast-burn family: percentage of minimum time (left) and of propellant at equal trip time (right) saved, against initial acceleration, for Mars and Jupiter at two launch energies (only table nodes where the simulation solver succeeded). The saving is almost entirely in time, not in propellant.

fixed, the closure probability is 0.74, and the failure mode is propellant mass, driven by the specific-mass range of 28.7–45.6 kg/kWe. This case uses the $C_3 = 0$ trajectory table, which has no failed nodes, so it is unaffected by the defect described in Section 4.3.

Era designs on the common inputs. Table 19 gives the mass ledger of the minimum-mass designs of the three missions in the three eras, computed with the common inputs and the optimal-control trajectory model (Section 4.3). Figure 6 shows why the era matters. The payload fraction at the optimum I_{sp} falls steeply with power-system specific mass, so in the Mars case the 2020s power system (about 80 kg/kWe at 100 kWe) leaves 0.6 of the launch mass for payload and bus, the 2030s system (42 kg/kWe) leaves 0.72 and the 2040s system (27 kg/kWe) leaves 0.78. For Jupiter, the corresponding fractions are 0.41, 0.56 and 0.63. Figure 7 shows how the optimum I_{sp} itself moves with the era.

4.5 Thermal and radiator design

Every kilowatt of electric power leaves a fission system with four to ten kilowatts of heat that can only be radiated, so the radiator, not the reactor, sets the geometry of the vehicle. The SMAD study sizes it with the emitted flux $q = \varepsilon\sigma F\eta_{fin}(T_r^4 - T_s^4)$ per emitting surface and $A = 1.10\,Q_{rej}/q$, where the 10 % margin

Table 8: Reconciliation: P_e [kWe] / escape or launch mass [t] / trip time [d] of the minimum-mass design in each model. "none" = no closure within the power grid (8–600 kWe); "n/a" = no table.

Mission	Era	SMAD native	sim native	common + dV(T) law	common + burn-coast-burn	common + optimal control
Mars cargo 8 t, 450 d	2020s	96 / 28.0 / 427	63 / 21.4 / 402	81 / 23.2 / 427	none	95 / 24.9 / 428
	2030s	60 / 17.2 / 427	49 / 15.3 / 396	57 / 16.2 / 428	none	64 / 16.9 / 428
	2040s	62 / 15.0 / 427	43 / 13.8 / 400	60 / 14.4 / 427	none	80 / 14.9 / 428
Jupiter orbiter 1.5 t, 6 yr	2020s	23 / 9.7 / 2082	72 / 17.7 / 1866	22 / 8.7 / 2086	86 / 19.3 / 2081	89 / 20.4 / 2082
	2030s	18 / 6.1 / 2082	38 / 8.9 / 1882	17 / 5.7 / 2084	52 / 9.6 / 1809	45 / 9.4 / 2082
	2040s	15 / 5.2 / 2082	33 / 7.4 / 1819	16 / 5.1 / 2082	52 / 8.1 / 1627	40 / 7.6 / 2082
Uranus orbiter 1 t, 10 yr	2020s	26 / 11.7 / 3469	none	23 / 9.7 / 3475	none	none
	2030s	17 / 6.2 / 3469	none	16 / 5.8 / 3472	none	105 / 17.9 / 3469
	2040s	15 / 5.4 / 3469	none	15 / 5.1 / 3465	none	85 / 11.2 / 3469

Table 9: Values Changed from the Corrected Paper Case (B) to the Redesign (C) (source: `sim/changes_log.md`).

Item	Old (B)	New (C)	Reason
Trip time	200 d (dV 6.25 km/s, below the 259-d Hohmann time)	492 d, full integration incl. capture	the 200-d definition has no source
Thruster	Hall 2000 s, η 0.50	NEXT 4170 s, η 0.706, 9 units	life 51,184 h per unit [6]; count sized for $\leq$80 % life
Power	100 kWe reference	71.9 kWe	minimum initial mass under the policy
f_{EP}	0.5	0.93 ($= 1 - 5\,\mathrm{kWe}/P_e$)	half the power on the bus is not credible at tens of kWe
Reserve	none	15 % on bus, power, thrusters	early-design growth (assumption)
Isp	2000 s	4170 s; optimum of the generic fit 3700 s	power-limited trade, flat to 0.1 % over 3600–4170 s
Escape mass	9.5 t	11.8 t at $C_3 = 0$	output of the closure

covers heat-pipe and meteoroid losses and the area counts both faces of two-sided panels, as in [11, 13]. The study uses $\varepsilon = 0.85$, $F = 0.95$ and $\eta_{fin} = 0.90$, with a sink temperature for a flat panel within 30° of the Sun line of $T_s^4 = \alpha S(r)\sin\beta/(2\varepsilon\sigma) + T_{cmb}^4$ ($\alpha = 0.25$ at end of life). The sink temperature is 205 K at 1 AU, 90 K at Jupiter, 47 K at Uranus and 37 K at Neptune. As a result, the design is set by the hottest case (1 AU, early in the mission), and the outer planets only add margin in the cold case.

Temperature trade. Raising the radiator temperature shrinks the area as $1/(T_r^4 - T_s^4)$, but it also lowers the cycle efficiency $\eta_{cyc} = f_C(1 - T_r/T_{hot})$, which inflates the reactor, shield and fuel. Figure 8 shows the result for the 17.5 kWe Jupiter orbiter. There is a shallow interior minimum of power-system mass at about 550 K for Brayton, Stirling and AMTEC conversion (1346, 1315 and 1274 kg) and at about 790 K for silicon-germanium thermoelectrics (1783 kg), with radiator areas of 27, 19, 14 and 76 m^2 at the respective minima. For an ideal Carnot machine with a fixed-size radiator the optimum would be $0.75\,T_{hot}$ (862 K for 1150 K). The lower real optimum reflects the Carnot fraction $f_C < 1$ and the step in radiator areal density from 3.8 kg/m^2 (composite panels with water heat pipes, 375–550 K, COMPASS [11]) to 7.67 kg/m^2 for armoured refractory heat pipes above 650 K [8]. A Brayton cycle optimised for 540 K, as in [13], is an input assumption, and holding the Carnot fraction constant as T_r rises is optimistic.

Radiator numbers per case. The advanced-tier designs radiate 307 kWt (Mars cargo, 98 m^2 at 540 K, 504 kg), 90 kWt (Jupiter, 29 m^2, 147 kg) and 86 kWt (Uranus, 27 m^2, 140 kg). The near-term Mars stack needs 956 kWt at 720 K (96 m^2, 1342 kg) because its power-system efficiency is lower. All of these are far below the 2,500 m^2 of the largest studied system [11]. Since the area does not exceed 10^2 m^2 in any case, the issue is deployment rather than area: panels of 4×5 m form two wings of 24–34 m inside the shield umbra. The areal densities used are anchored to 3.8 kg/m^2 (COMPASS), 8.8 kg/m^2 (JIMO Brayton, [13]) and 7.67 kg/m^2 [8], with a 0.55–1.6 multiplier covering the spread down to the 2.2 kg/m^2 carbon-carbon goal [15], which is a development goal and not a measurement.

Bus thermal control. The PPU waste heat (4.0 kW for Mars, 0.8 kW for the outer planets) is rejected on a 330 K cold-plate radiator (26 kg at Mars). MLI at 0.73 kg/m^2 and 2–5 % of the dry mass for the thermal subsystem follow SMAD [5]. Radiation from the hot wings onto the bus is only about 1 W, because the wings are edge-on; a 10° tilt raises it by 10–30×, which is still only tens of watts. In the outer solar system, a cold bus is cheap to keep warm with a fission source, since the kilowatt-class electric budget dwarfs the tens-of-watt MLI leak; this is an advantage over RTG-powered probes. This part of the analysis is at the differential-element level, and a full thermal model is still needed.

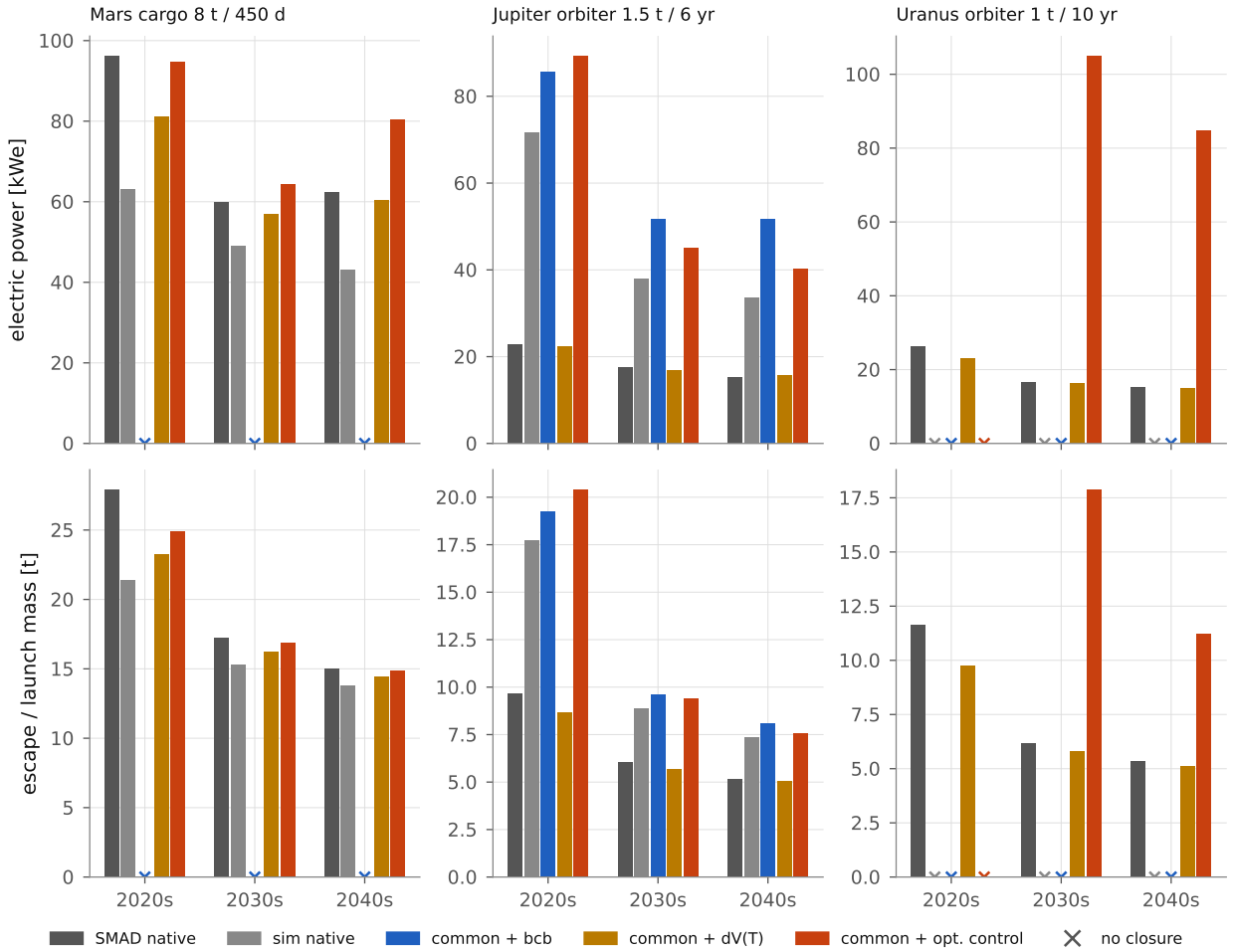


Figure 4: Power (top) and mass (bottom) of the minimum-mass designs in the SMAD study, the simulation, and the common closure with three trajectory models. Crosses: no closure (the burn-coast-burn family has none for Mars; the simulation's own Mars bars are artefacts, E-S0). For Mars the common closure and SMAD agree; for Jupiter they differ through the capture definition.

4.6 SMAD subsystem sizing and technology readiness

The SMAD-style study sizes each vehicle bottom-up, following Larson and Wertz [5]: requirements lead to a functional breakdown, then to sizing equations, and finally to mass and power budgets with maturity-based margins. The 2011 edition was not available, and nothing is taken from it. The margins are a 25 % mass-growth allowance on new equipment (reactor, converter, PMAD, radiator, shield, boom, harness), 15 % on the PPU, feed, structure, ADCS, communications, C&DH, thermal and auxiliary power (an interpolation inside the SMAD range), 5 % on flown thrusters and tanks, 2 % for integration and a 10 % system margin. The usual pre-Phase-A system margin in concurrent engineering is 30 % [16]; the lower value is kept because the item margins are already large. Power loads carry 20 %, ΔV carries 5 %, and a 15 % launch reserve is applied. Structure is at least 8 % of the injected mass and thermal control at least 2 % of the dry mass (SMAD Table 10-10).

Subsystems. The *reactor* mass is $m = 436 + 0.1195\,P_{th}$ kg (fitted to AMTEC-SNRP and SP-100) with a 25 % margin; the burnup is 0.2–0.4 % against a 3 % limit, so the fuel is not a constraint. The baseline *conversion* is a closed Brayton cycle (He-Xe, 1150 K inlet, 21.7 % cycle, 7.2 kg/kWe converter, TRL 5, about 50,000 h of ground test), with Stirling, AMTEC and SiGe thermoelectrics as alternatives; the last is the only flight-proven conversion, but it costs 13–17 % more launch mass. The *shield* is a LiH/W cone sized to a dose limit of 25 krad(Si) and 10^{11} n/cm^2 at the payload over the mission. It comes from a point-source exponential-removal model that reproduces the JIMO shield to within a factor of 1.9 (conservative), and the shield half-angle and boom length are optimised jointly (2–3°, 31–41 m). For *propulsion*, NEXT-class ion thrusters are used (6.9 kW, 13.5 kg assumed) with a spare count of $\max(2, 15\,\%)$, a PPU of 5 kg/kW sized 1.15× for a spare string, and xenon tanks at about 10 % of the propellant (carbon-overwrapped; krypton and argon are heavier and lose efficiency). The *structure, ADCS, communications and C&DH* comprise a

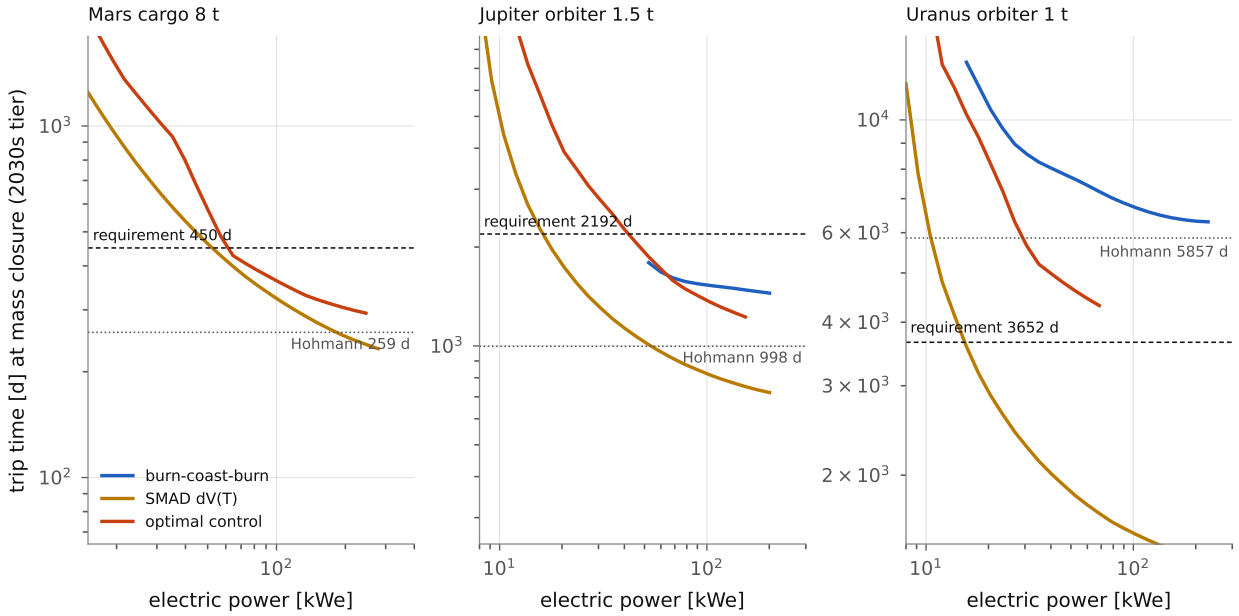


Figure 5: Trip time versus electric power at mass closure (2030s tier, thrust-always-on fastest solution for the optimal-control curve), with the requirement and the Hohmann flight time. Where the burn-coast-burn curve is absent, the family has no solution.

Table 10: Mass ledger [kg] of the minimum-mass optimal-control designs (common inputs); radiator in m^2, thruster life as fraction of the demonstrated NEXT value, times in days.

Mission	Era	P_e	m_{ps}	m_{thr}	m_{bus}	reserve	tanks	Xe	m_0	A_{rad}	N_{thr}	life	T_{helio} / T_{cap}
Mars cargo 8 t, 450 d	2020s	95	7685	747	3278	1171	366	3655	24901	94	13	0.21	363 / 65
	2030s	64	2935	511	2147	559	248	2475	16875	106	9	0.21	363 / 65
	2040s	80	2276	633	1966	487	140	1396	14898	44	11	0.14	370 / 57
Jupiter orbiter 1.5 t, 6 yr	2020s	89	7594	681	1564	984	737	7370	20429	191	12	0.78	1474 / 609
	2030s	45	2321	327	1160	381	338	3381	9408	74	6	0.77	1478 / 605
	2040s	40	1483	280	1133	290	262	2617	7564	29	5	0.68	1517 / 566
	2030s	105	4544	855	1162	656	880	8804	17902	173	16	0.79	3331 / 138
	2040s	85	2500	638	1138	428	501	5010	11214	61	11	0.64	3351 / 119

30–45 m boom with a first bending mode ≥0.1 Hz, thrust-offset torque unloaded by thruster gimbals, a 4 m Ka-band antenna and rad-tolerant processors.

Technology readiness. Table 11 summarises the readiness of the critical items today. The reactor, converter, radiator, shield, PMAD and autonomous operation all sit at TRL 4–5, and only the thrusters, xenon storage and C&DH are at TRL 6 or above. The binding quantity is the converter life: a full-power life of 3, 8 and 12 yr is 0.5, 1.4 and 2.1× the 50,000-h demonstration.

Mass budgets. Figure 9 shows the SMAD launch-mass breakdown by case and era. The power system is 20–45 % of the dry mass and shrinks from 8 t (2020s Mars) to 2 t (2040s), the propellant is 13–33 % of the launch mass, and the margins plus integration are 8–10 %.

Development plan by tier. The 2030s state needs a flight-like reactor and shield ground test, a full-scale deployed radiator, a 100-kWe-class converter and, as the critical path, a 100,000-h (11.4-yr) real-time converter life test starting in 2029. The flight unit is then a 2044 article, or a 2038 article if 50,000 h is accepted, at the cost of a 1.4–2.1× gap for the outer-planet missions. The 2040s state adds a 1500-K Brayton with a new fuel form and refractory alloys, a 2.2 kg/m^2 carbon-fibre radiator, 100 krad / 10^{13} n/cm^2 avionics qualification, a 30,000-h high-I_{sp} thruster test and an operational refuelled Starship. Its critical path is again the converter (150,000 h of partly accelerated evidence), and the flight unit is a 2050 article. Figure 10 gives both schedules; the durations are assumptions except where a test length is fixed by physics. The tier names (2020s, 2030s, 2040s) are technology *states*, not forecasts: they state what must be true, and when it could be true if funded from 2027.

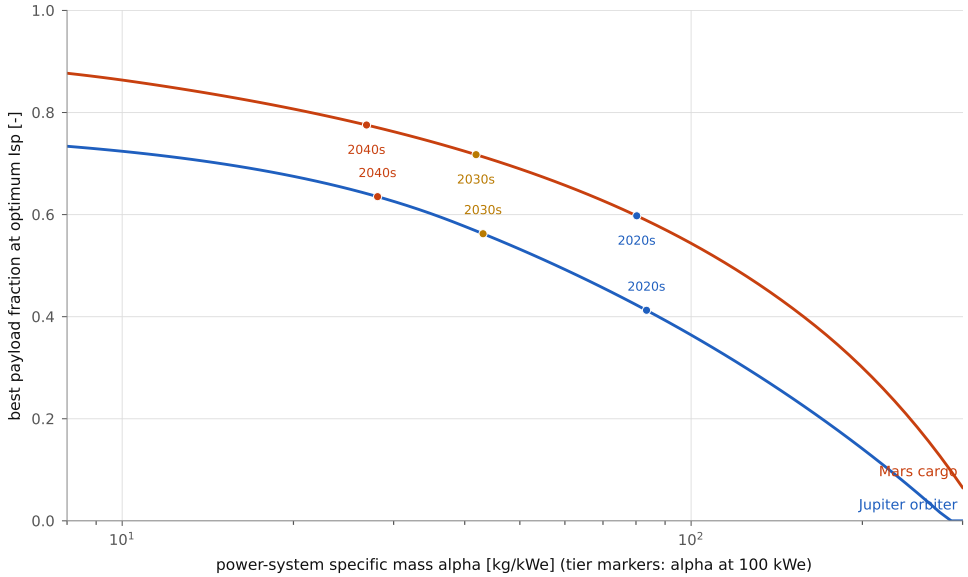


Figure 6: Best payload-plus-bus fraction (at the optimum specific impulse) versus power-system specific mass. Markers: the specific mass at 100 kWe in each era. A lighter power system leaves a markedly larger fraction for payload and bus.

Table 11: Readiness of the Critical Items in 2026 (SMAD study; the TRL values are our judgement from the cited histories).

Item	TRL	Demonstrated vs missing
Fission reactor, 100–500 kWt, years at full power	4	KRUSTY 1 kWe ground test 2018 [17]; SP-100 ground tests, never flown [8]. Missing: a flight-qualified reactor with multi-year fuel qualification.
Brayton conversion	4–5	10 kWe Brayton unit about 50,000 h on the ground; none has run in space [13, 14]. Missing: 100 kWe unit and a 100,000-h test.
Heat-rejection, 30–100 m^2 at 540 K	4	Sub-scale and full-scale panels, NaK loops to 875 K [14]. Missing: deployment of 100s of m^2 in vacuum.
Shield (LiH/W)	4	Ground-test history; LiH swelling and hydrogen containment problems found under Prometheus. Missing: flight-like thermal management.
PMAD ∼1 kV and direct drive	4–5	960 Vac concept [14]; direct-drive Hall units TRL 3–4 [18, 19].
Thrusters (NEXT-class ion)	6	51,184 h and 918 kg Xe [6]; HERMeS 3,570 h wear test [20].
Boom, ADCS, nuclear launch approval	5	Analysed, not flown; US launched only SNAP-10A.

4.7 Operations-level design: power flow, data, control, reliability, operations and cost

The sizing of Section 4.6 closes the vehicle in mass and power. SMAD's design process also asks what the mission is for, how its data get home, how it is controlled and operated, how likely it is to survive and what it costs [5]. This subsection adds those analyses for the 2030s designs. Unless stated otherwise the numbers are computed by `v3/framework.py` from the SMAD design points and the common closure, and every assumption is marked (A).

4.7.1 How the power system works

Figure 11 traces the power of the 2030s Mars design from the reactor to the jet. Of 372 kWt of reactor heat, the Brayton converter turns 19.8 % into 73.5 kW of alternator output, and the main radiator rejects the remaining 299 kW (80 % of the reactor power). PMAD, alternator and parasitic losses take 13.6 kW, the bus keeps 2.2 kWe, and the PPUs receive 57.7 kW, of which 53.7 kW reaches the thrusters and 34.9 kW leaves as jet power, giving 2.18 N at 3270 s. Only 9 % of the reactor heat therefore becomes thrust power, and every step of the chain is a heat load that a radiator must reject. Figure 12 shows the radiator side of the same balance. The rejected heat per square metre of emitting area follows $q = \epsilon\sigma F\eta_{fin}(T_r^4 - T_s^4)/1.10$ with $\epsilon = 0.85$, $F = 0.95$, $\eta_{fin} = 0.90$ and a 10 % area margin (SMAD study, section 3.3). The 2030s designs sit at 540 K and 3.0 kW/m^2, so the Mars design needs 98 m^2; the 2040s designs run at 640–700 K, reject

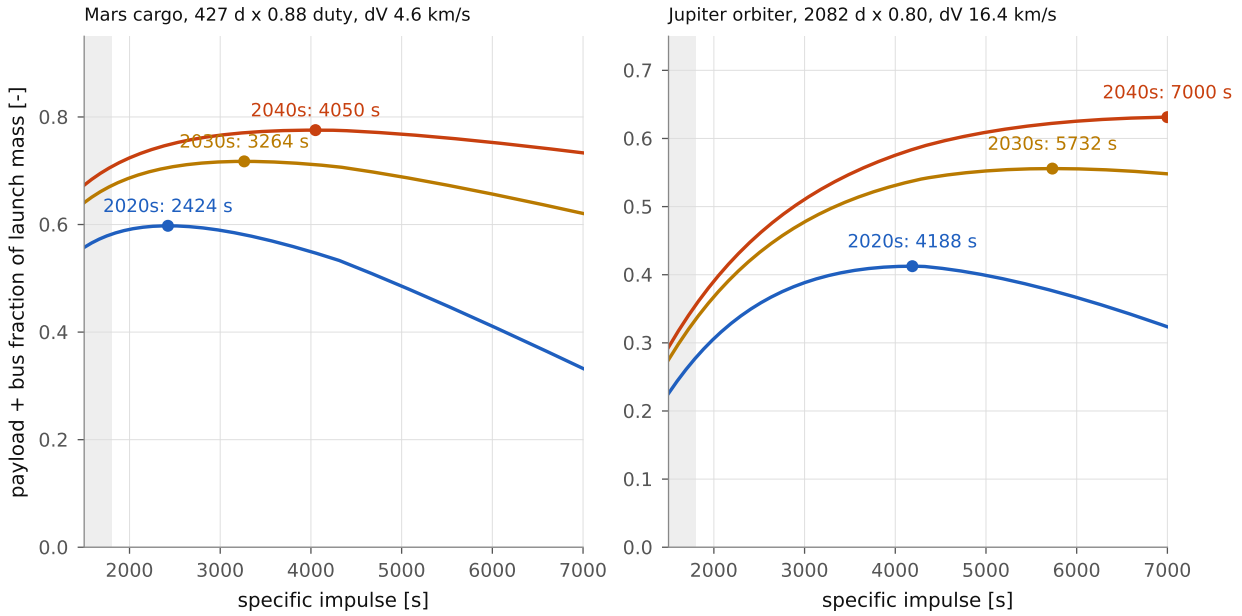


Figure 7: Payload-plus-bus fraction versus specific impulse in three eras for the Mars and Jupiter cases (power-limited model with the fitted thruster efficiency; above 4300 s the efficiency is held at its 4300 s value, so the 2040s Jupiter optimum at 7000 s is an extrapolation). Dots mark the optimum.

6.3–9.1 kW/m^2 and need about a third of the area (34 m^2 for Mars). Because q grows with T_r^4 while the Carnot efficiency falls with T_r, the radiator temperature is set by minimum mass, not by efficiency. Figure 13 gives the mass and power budgets of the three 2030s SMAD design points. In mass, the payload dominates the Mars stack (8.0 of 17.2 t) and the power system and bus make up more than half of the orbiters' dry mass (53 and 58 %). In power, the thrusters take 96 % of the Mars electric power but only 63–65 % of the orbiter power, because the science payload is allocated 2–3 kWe and the heaters and communications another 1.3–1.7 kWe.

4.7.2 What the mission is for, and how its data get home

The Mars cargo exists to deliver mass, and the two orbiters exist to return science data, so the communications link is part of their purpose rather than a support item. SMAD sizes a link through $E_b/N_0 = PL_l G_t L_s L_a G_r/(kT_s R)$ (Eq. 13-4, PDF p. 285) and a link budget with margin (PDF p. 293); with the SMAD study's budget (Table 12 note) our implementation reproduces its 60.2 kbps at Uranus. Table 12 and Fig. 14 compare four link options. A NEP vehicle can afford a 2 kW Ka-band transmitter (A), which at an assumed 50 % DC-to-RF efficiency is 9 % of the Jupiter and 4 % of the Uranus electric power. It raises the Uranus rate from 60 to 602 kbps on a 34 m antenna (2550 kbps on a 70 m), the Jupiter rate to 6.9 Mbps and the Mars rate at maximum distance to 37 Mbps. The Psyche DSOC experiment returned 25 Mbps optically at 1.5 AU, against a 1 Mbps goal [21], and 267 Mbps at 31 million km [22]; scaled by $1/d^2$ with the same 22 cm terminal (A), it would give 128 kbps at Uranus, which a NEP power budget could raise by a larger laser. The data volume is the practical measure. The UOP concept returns 51.9 GB at 19.5 kbps [23], which needs 5915 h of 34 m antenna time; the NEP link needs 192 h. This matters because DSN demand already exceeds supply by up to 40 %, rising to about 50 % by the 2030s [24]. At Mars, MAVEN's relay ended in 2026 [25], and a NEP cargo tug left in Mars orbit with 64 kWe and a 2 kW Ka link would be a relay with ten times the rate of the 200 W Ka-band baseline. This is the clearest non-propulsive use of the reactor.

Command and data handling. The C&DH sizing of the SMAD study is a dual-string unit (35 kg base after the NASA/AMA mass-equipment list, plus remote interface units along the boom and box shielding, 90 W) with a 256-Gb (32 GB) recorder (A). At the NEP Uranus rate the recorder empties in about 15 eight-hour passes (2.2 GB per pass), and at the baseline rate in about 150, so the recorder, not the radio, would limit high-rate instruments in the baseline design. The electronics near the reactor see the dominant radiation dose: SMAD lists commercial parts at 1–10 krad and rad-hard parts at 100 krad–1 Mrad (Table 8-7, PDF p. 126), and the boom electronics are therefore placed in the shield umbra and use hardened parts.

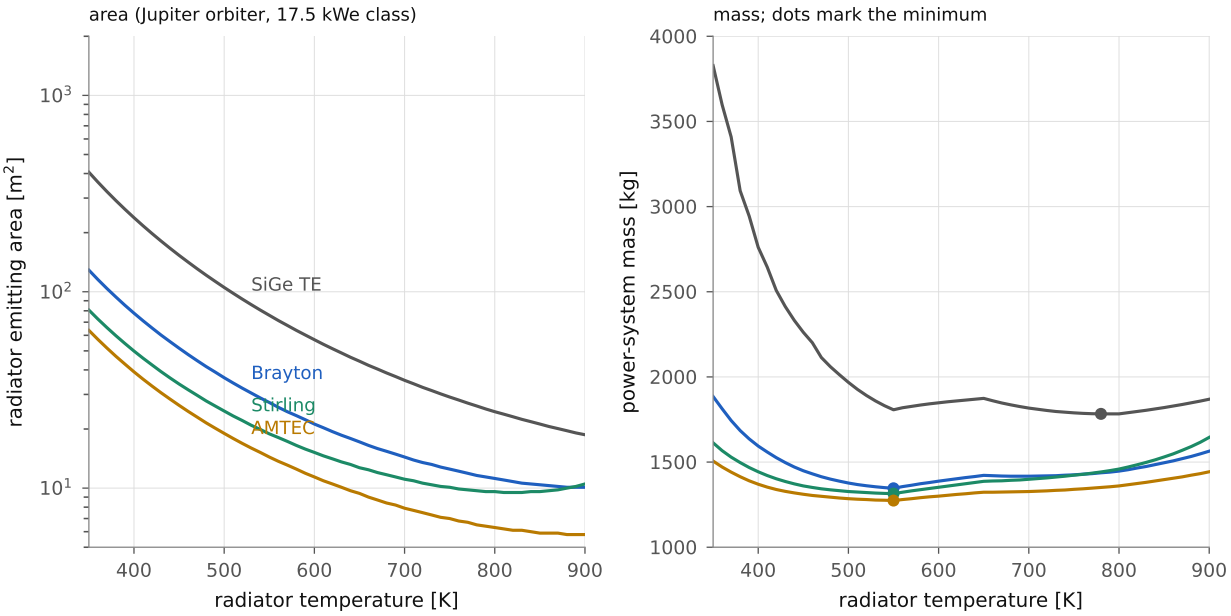


Figure 8: Radiator emitting area (left) and total power-system mass (right) versus radiator temperature for four conversion technologies at the Jupiter-orbiter design power (SMAD study). Dots: mass minimum. The mass minima are shallow for all four technologies.

Table 12: Downlink Data Rate by Link Option and Distance, and DSN Time to Return 51.9 GB from Uranus

Link	Mars (2.67 AU)	Jupiter (6.2 AU)	Uranus (21 AU)	DSN hours for 51.9 GB
Ka 2 kW RF, 4 m (NEP power, A)	37.2 Mbps	6.9 Mbps	602 kbps	192
Ka 2 kW RF, 4 m, 70 m DSN (A)	157.7 Mbps	29.3 Mbps	2.5 Mbps	45
Ka 200 W RF, 4 m (SMAD baseline)	3.7 Mbps	690 kbps	60 kbps	1917
Optical, DSOC terminal scaled $1/d^2$ (A)	7.9 Mbps	1.5 Mbps	128 kbps	904
X 50 W RF, 4 m (safe-mode)	152 kbps	28 kbps	2.5 kbps	47032
UOP Ka-band, 3.1 m HGA (cited [23])	–	–	19.5 kbps	5915

Link equation SMAD Eq. (13-4) [5] with the SMAD study's budget (antenna efficiency 0.55, T_{sys} 40 K Ka / 25 K X, E_b/N_0 5 dB, 2.8 dB line, pointing and polarisation losses, 2 dB Ka weather, 4 dB margin); it reproduces the study's 60.2 kbps at Uranus. Optical row scales the DSOC 25 Mbps at 1.5 AU [21] by $1/d^2$ (A). 51.9 GB is the UOP science return [23].

Attitude control with a long boom. The SMAD study sizes the ADCS for a reactor 25–45 m from the payload. The dominant disturbance is the thrust-vector offset (5 mm residual after gimbal trim gives 0.011 N m for Mars), far above solar pressure, so four 25 N m s wheels are unloaded by the thrusters every hour. The boom's first bending mode is about 0.1 Hz, and the control bandwidth is held at one tenth of it (0.01 Hz). SMAD warns that long deployed booms are sensitive to thermal shocks when entering or leaving eclipse (PDF p. 190), which favours a truss boom and slow attitude manoeuvres. A 50-mm offset, for example from untrimmed xenon migration, raises the Mars torque tenfold and forces continuous gimbal trimming.

4.7.3 Reliability and redundancy over a decade

SMAD defines mission reliability through $R = e^{-\lambda t}$ for an item without wear-out and k-of-n redundancy for parallel items (Eqs. 19-1 to 19-6, PDF pp. 392–393), and notes that dormant units fail at about a tenth of the active rate (PDF p. 393). The difficulty for NEP is not the formula but the evidence. With T failure-free unit-hours of testing, the most that can be claimed at 60 % confidence is $\lambda \leq 0.916/T$. Table 13 and Fig. 15 apply this to the thrusters and converters of the 2030s designs. With one NEXT long-duration test (51,184 h [6]) as the only evidence, the thruster strings of the closure designs succeed with a demonstrable probability of only 0.18 (Mars) and 0.01 (Jupiter); 3, 6 and 13 spare thrusters would be needed for 0.9. If ten such units had been run (A), one or two spares would suffice. For the converter, even the 100,000-h test of the 2030s plan demonstrates a single-unit reliability of only 0.53 over the Jupiter mission and 0.38 over the Uranus mission. To demonstrate 0.9 at 60 % confidence the programme needs 8.7 times the mission hours in failure-free converter testing: 229, 610 and 915 thousand unit-hours, i.e. several converters run in parallel for the full mission life. This is a cost item (parallel test units) rather than a schedule item, and it is the strongest reliability argument for 2×100 % converter redundancy, which raises the demonstrable

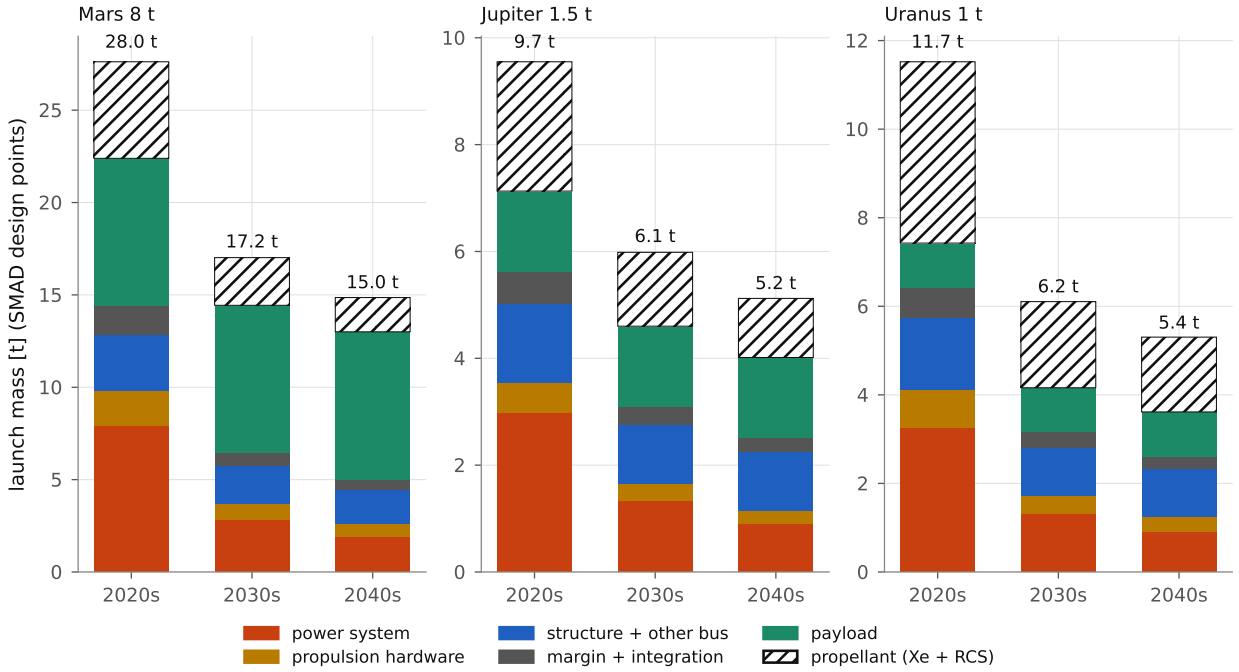


Figure 9: Launch-mass breakdown of the three SMAD design cases in the three technology eras (SMAD native design points, predicted masses with margins). Labels give total launch mass.

Table 13: Reliability the Test Evidence Can Demonstrate (60 % Confidence, Zero-Failure Bound)

Mission (2030s design)	Thrusters active / total	Burn [h]	P(string), 51 kh evidence / spares for 0.9	same, 10 units (A)	Converter hours	R one / 2×100 %, 50 kh	same, 100 kh	Test unit-kh for R = 0.9
Mars cargo 8 t, 450 d	9 / 9	10715	0.18 / 3	0.84 / 1	26298	0.62 / 0.85	0.79 / 0.95	229
Jupiter orbiter 1.5 t, 6 yr	6 / 6	39406	0.01 / 6	0.65 / 1	70128	0.28 / 0.48	0.53 / 0.78	610
Uranus orbiter 1 t, 10 yr	14 / 16	46154	0.00 / 13	0.90 / 2	105192	0.15 / 0.27	0.38 / 0.62	915

$\lambda \leq 0.916/T$ for T failure-free unit-hours (60 % confidence); $R = e^{-\lambda t}$ and k-of-n redundancy, SMAD Eqs. (19-1)–(19-6) [5]; dormant spares at one tenth of the active rate (SMAD PDF p. 393). Thruster evidence: NEXT 51,184 h [6]; converter evidence: about 50,000 h Brayton [14].

full-power probability for Jupiter from 0.53 to 0.78 with the 100,000-h test.

4.7.4 Deployment sequence, nuclear safety and mission timeline

Table 14 sets out the operations sequence from launch to end of life, and Fig. 16 places it on a calendar together with the development gates of Section 4.8. The vehicle is launched with the reactor never critical: SMAD records that SNAP-10A was launched subcritical and designed to remain subcritical on impact (PDF p. 428). The boom and radiators are deployed and their flexible modes identified before start-up, because a reactor that has run cannot be approached. The SMAD study keeps the start-up until escape or a nuclear-safe orbit; SR-1 Freedom plans to start its reactor within 48 h of launch [27], which is the more aggressive precedent. EP checkout follows thruster by thruster, since the NEXT-C anomaly on DART was traced to insufficient electrical testing of the thruster and spacecraft together [28]. The heliocentric and capture phases follow the optimal-control profile, the reactor's remaining full-power life is spent on delivery or science operations (about 1.8, 2.3 and 2.5 yr for Mars, Jupiter and Uranus), and at end of life the reactor is shut down and the vehicle left in a long-lived orbit, as SNAP-10A still is [29]. Launch approval runs in parallel with development; SMAD gives 1–2 yr for an environmental impact statement and up to 5 yr with international concerns (PDF p. 427, as used in the SMAD plan).

4.7.5 Cost estimate

SMAD's parametric cost-estimating relationships are valid only within 25 % of their data ranges (PDF p. 407), and the NEP vehicles here (5–25 t) are far outside them; the SMAD study therefore used an analogy to JIMO. We replace it with an explicit two-part model. The non-power spacecraft (bus, thrusters, tanks, contingency and, for the orbiters, the science payload) is costed with a power law fitted to five deep-space missions with published life-cycle cost and dry mass (Dawn, Psyche, Juno, the UOP study and Europa Clipper; exponent 1.59, log standard error 0.27). The nuclear power system is an adder of \$0.5–2.5 B (log-uniform, A), bracketed by the \$0.499 B DRACO flight-demonstrator agreement [30] and by NASA's more than \$20 B spent on more than a dozen attempted space-reactor programmes [27], widened by SMAD's

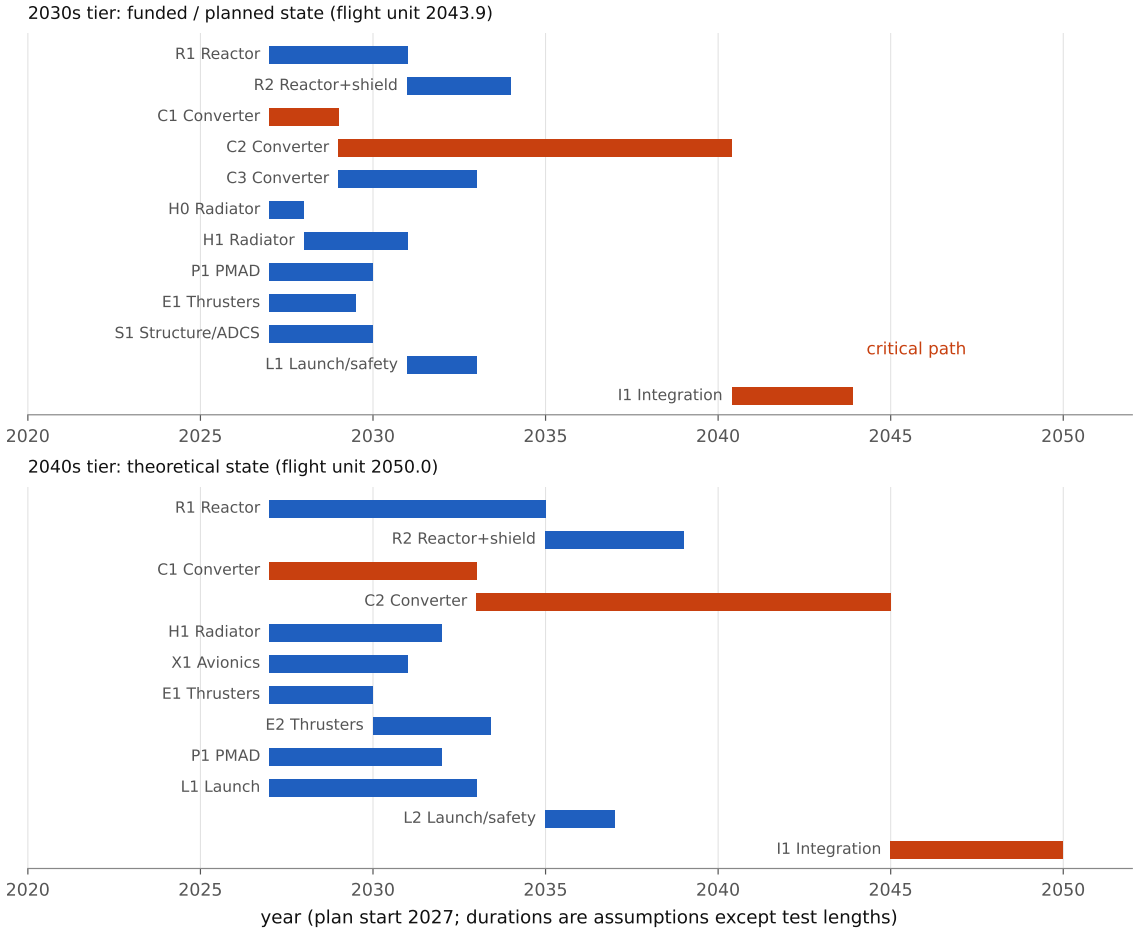


Figure 10: Development schedules implied by each tier (infinite-resource earliest finish, plan start 2027). Red: critical path, which is the converter life test in both tiers.

technology-readiness cost uncertainty (Table 20-15, PDF p. 412). Table 15 and Fig. 17 give the result. The 2030s Mars cargo costs \$5.7 B (P10–P90 4.1–7.9), the Jupiter orbiter \$6.2 B (4.5–8.6) and the Uranus orbiter \$8.1 B (5.9–11.1) in the 2030s state or \$6.2 B in the 2040s state. These are flagship costs, comparable with Europa Clipper (\$5.2 B [31]) and above both UOP estimates. The estimate is low-confidence: the analogy set is small and mixed in dollar year, the 2020s designs are extrapolated beyond it, the power-system adder is an assumption, and a launch on SLS (\$4 B per flight [32]) or a Starship tanker campaign is not included. Recent flagship experience (Dragonfly at about twice its proposed cost [33]) suggests that the risk is on the high side.

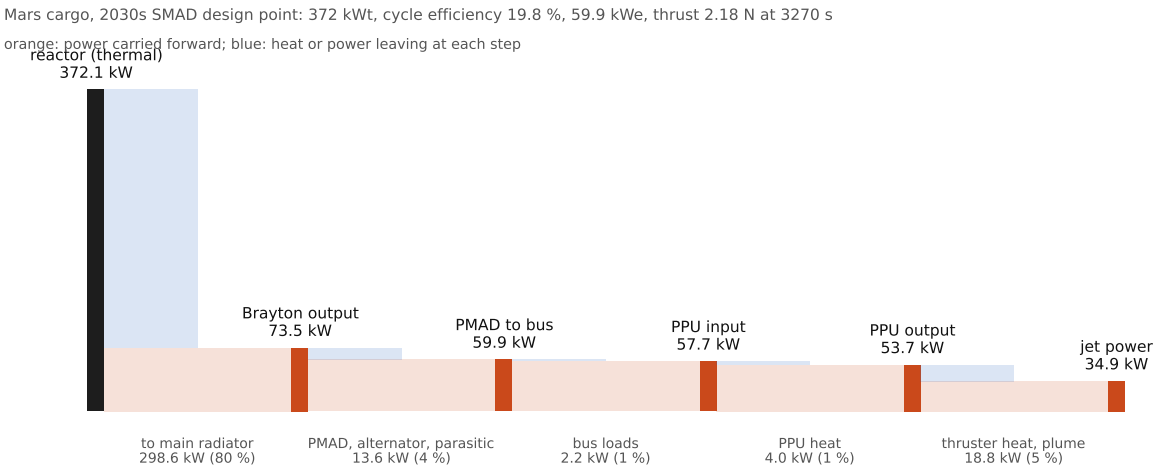


Figure 11: Power flow of the 2030s Mars cargo design from reactor heat to jet power, with the power leaving the chain at each step. The radiator, not the thrusters, receives most of the reactor's output. SMAD design point (smad/design_points.json); PPU efficiency 0.93 and thruster efficiency 0.65 at 3270 s.

Table 14: Deployment and Operations Sequence from Launch to End of Life

#	Phase	Duration	Basis
1	Launch, separation, checkout (reactor never critical, launched subcritical with locked control drums)	7 d (A)	SMAD PDF p. 428: SNAP-10A was launched subcritical and designed to stay subcritical on impact
2	Boom and radiator deployment, flexible-mode identification	14 d (A)	first mode about 0.1 Hz, control bandwidth $\leq f_1/10$ (SMAD study 3.6); boom thermal shock on eclipse exit, SMAD PDF p. 190
3	Reactor start-up and power ascension after reaching escape or a nuclear-safe orbit	30 d (A)	SR-1 plans start-up within 48 h of launch [27]; SNAP-10A used about 1300 km (SMAD study 3.9)
4	EP checkout, thruster-by-thruster, PPU and PMAD at power	30 d (A)	DART's NEXT-C anomaly came from insufficient thruster/spacecraft electrical testing [28]
5	Heliocentric thrust and coast (optimal-control profile)	Mars 363 d, Jupiter 1478 d, Uranus 3351 d	computed (common closure)
6	Capture spiral to the delivery or science orbit	Mars 65 d, Jupiter 605 d, Uranus 119 d	computed
7	Delivery / science operations, relay at high power	rest of the full-power life (3, 8, 12 yr)	reactor life from the mission set; data volume Table 12
8	Disposal: reactor shut down, vehicle left in a long-lived high orbit or heliocentric orbit; no Earth return	–	SNAP-10A still in a long-lived orbit [29]

Table 15: Life-Cycle Cost Estimate: Analogy Model for the Non-Power Spacecraft Plus a Nuclear Power-System Adder

Mission	State	Non-power dry mass [kg]	Spacecraft P50 [$B]	Power system P50 [$B]	σ_{ps}	Total P50 (P10–P90) [$B]
Mars cargo 8 t, 450 d	2020s	5561†	9.2	1.1	0.44	10.6 (7.7–14.6)
	2030s	3465	4.4	1.1	0.41	5.7 (4.1–7.9)
	2040s	3225	3.9	1.1	0.42	5.2 (3.7–7.3)
Jupiter orbiter 1.5 t, 6 yr	2020s	5465†	9.0	1.1	0.44	10.4 (7.5–14.3)
	2030s	3706	4.9	1.1	0.41	6.2 (4.5–8.6)
	2040s	3465	4.4	1.1	0.42	5.7 (4.1–7.9)
Uranus orbiter 1 t, 10 yr	2030s	4554†	6.7	1.1	0.41	8.1 (5.9–11.1)
	2040s	3704	4.9	1.1	0.42	6.2 (4.5–8.5)

Spacecraft: $C = e^{-11.47} m^{1.59}$ ($B, m in kg), fitted to Dawn, Psyche, Juno, the UOP study and Europa Clipper (log standard error 0.27; valid to 4051 kg by SMAD's 25 % rule, PDF p. 407); † extrapolated. Power system: log-uniform $0.5–2.5 B (A), bracketed by DRACO's $0.499 B [30] and more than $20 B over more than twelve programmes [27], times a lognormal with $\sigma_{ps} = \sqrt{\sigma_{TRL}^2 + 0.40^2}$ (SMAD Table 20-15, PDF p. 412). 20,000 samples, seed 20251103. Mixed-year dollars; launch of SLS ($4 B per flight [32]) or a Starship tanker campaign is not added.

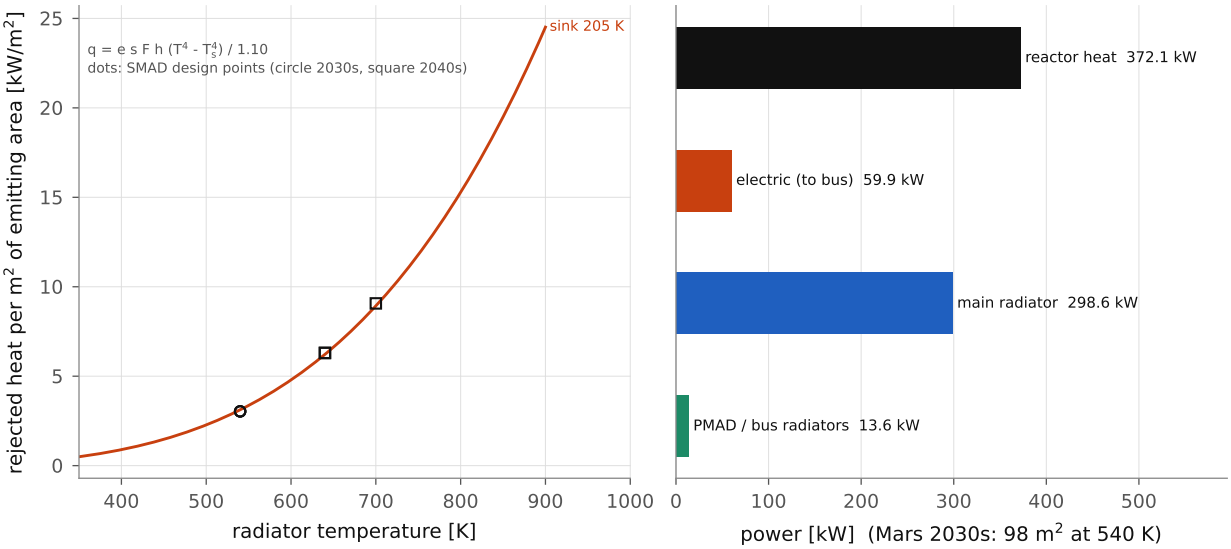


Figure 12: Left: radiator heat rejection per square metre of emitting area versus temperature, with the SMAD design points (circles 2030s, squares 2040s). Right: heat balance of the 2030s Mars design. Raising the radiator from 540 to 700 K triples the flux and cuts the area by two thirds. Radiator model and parameters of the SMAD study.

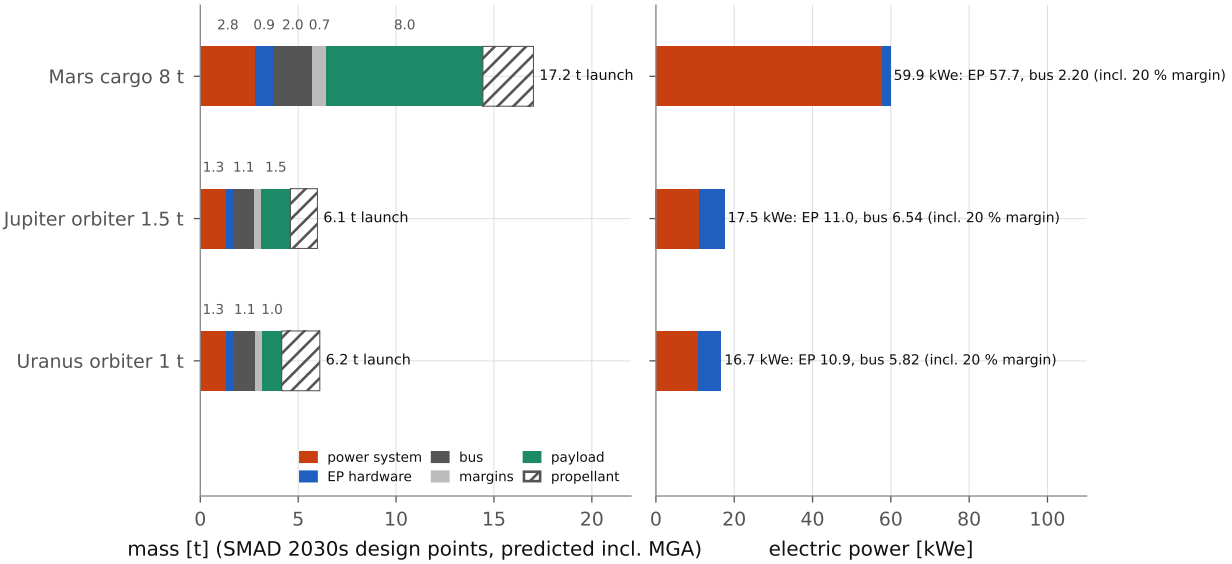


Figure 13: Mass budget (left, predicted masses with growth allowance; hatched: propellant) and electric power budget (right) of the 2030s SMAD design points. The orbiters spend a third of their power on the science payload and bus. Values from smad/design_points.json.

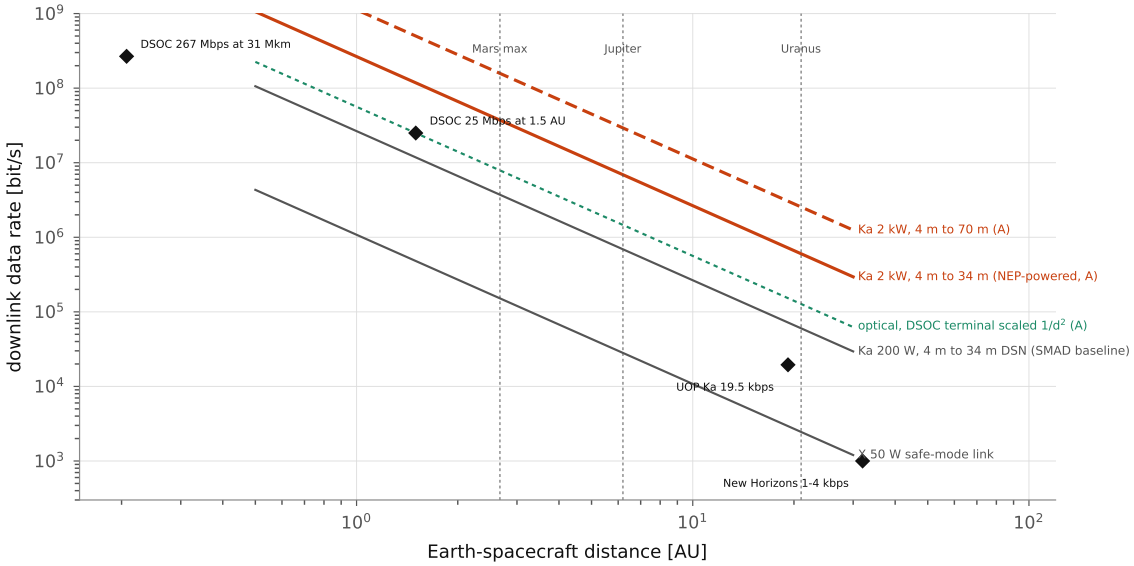


Figure 14: Downlink data rate versus Earth distance for Ka-band at the SMAD baseline (200 W) and at NEP-enabled power (2 kW, 34 and 70 m DSN), the X-band safe-mode link and a distance-scaled optical terminal, with flown and studied points. NEP power buys a factor of ten in rate at any distance. SMAD Eq. (13-4); markers from [21–23, 26].

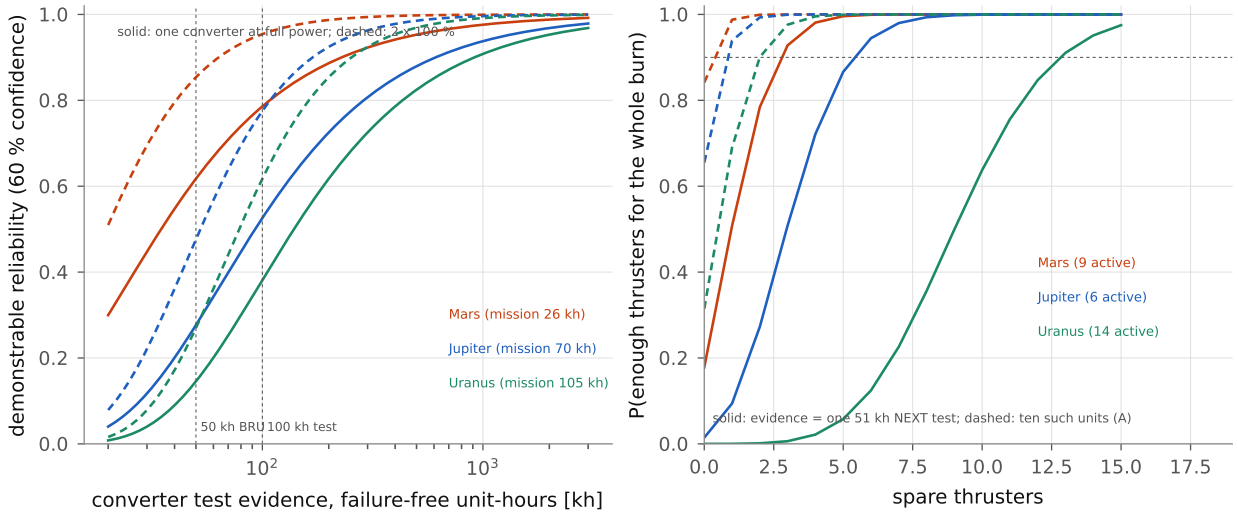


Figure 15: Left: converter reliability that a given amount of failure-free testing can demonstrate (60 % confidence) for each mission's full-power life, single unit (solid) and 2×100 % redundant (dashed). Right: probability that enough thrusters survive the whole burn versus the number of spares, for one NEXT-class test (solid) and ten (dashed, A). Evidence, not hardware, limits the reliability that can be claimed. SMAD Eqs. (19-1)–(19-6).

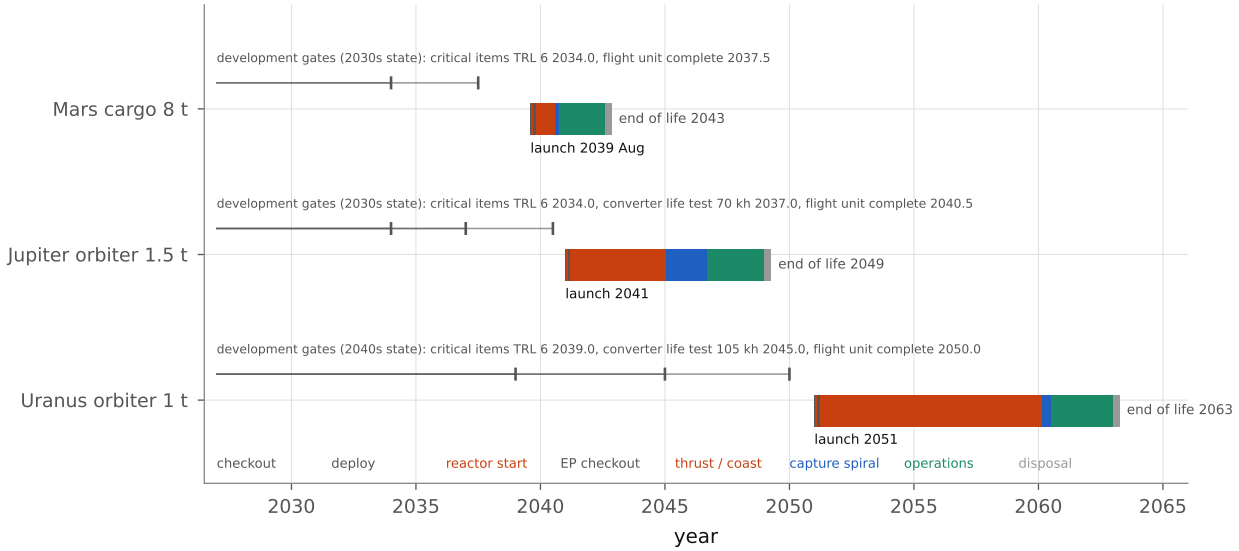


Figure 16: Mission timelines from development start to end of life for the earliest closing design of each mission: development gates (thin lines and ticks) and flight phases from launch to disposal. For the outer planets the converter life test, not the transfer, sets when the science starts. Values from v3/framework.py; phase durations before thrusting are assumptions (A).

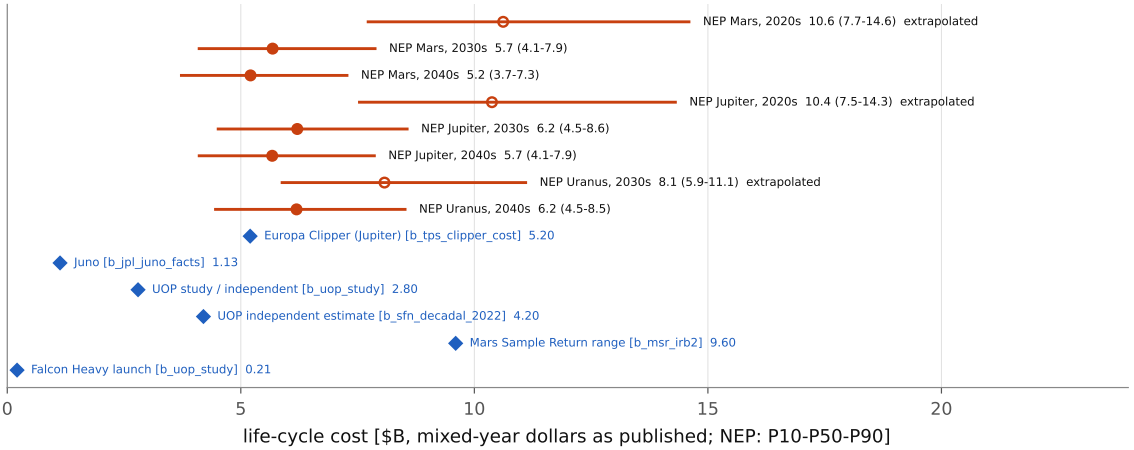


Figure 17: Estimated life-cycle cost of each NEP design (P10–P90 range and median; hollow: extrapolated beyond the analogy data) against published mission costs. Every NEP design is a flagship-class mission. Analogy model of Section 4.7.5; references in Table 21.

4.8 Mission results: technical closure and programmatic readiness

To decide which missions NEP can fly, and when, we separate two questions that the earlier version of this study had merged. The first is *technical closure*: does the vehicle close in mass, power, trip time and margins, and with what probability? The second is *programmatic readiness*: when could the technology state that closes it be qualified and flown, which items gate that date, and at what development cost and risk? The earlier single rule required a minimum critical TRL of 6 for a "possible" verdict. Because the 2040s technology state is newer, its critical items are at TRL 5 at the date of the state, and the lighter 2040s designs (Mars 14.9 t against 16.9 t, Jupiter 7.6 against 9.4 t, Uranus 11.2 against 17.9 t) were therefore ranked below the 2030s designs. A better technology state can always fly the hardware of an earlier one, so its verdict cannot be worse. The framework used here enforces that property by dominance and checks it in the code (`v3/framework.py`); for the three missions the raw closure classes are already monotonic.

Technical closure. The closure is computed with the common inputs and the optimal-control trajectory model, and is given as the power and launch mass of the minimum-mass design and the probability that some power closes the mission when the design is re-optimised in each Monte Carlo sample. A state *closes* a mission if this probability is at least 0.8 and closes *with low margin* if it is at least 0.5. The launch-vehicle margin and the sampled design power (5th–95th percentile) are reported beside it.

Programmatic readiness. For each state we compute the year in which its flight unit could be complete, as the latest of three gates plus integration. (1) The critical items (reactor and shield at power in a ground test, converter scale-up, radiator, PMAD) reach TRL 6: 2034 for the 2020s and 2030s states and 2039 for the 2040s state, from the development plans of the SMAD study. (2) The converter has run, in real time, for the mission's full-power life (26,000, 70,000 and 105,000 h for Mars, Jupiter and Uranus), starting when converter test articles exist (2029, or 2033 for the new 1500-K converter); the 50,000 h of Brayton ground test [13, 14] is accepted as covering missions that need less. (3) A launch vehicle exists for the stack: every NEP design here is heavier than one Falcon Heavy can inject, so SLS Block 1B or a refuelled Starship is needed. SLS is proposed for phase-out after three flights [32] and Starship ship-to-ship propellant transfer has not yet been demonstrated [34]; we assume refuelling is operational by 2030 (A). Integration takes 3.5 yr (5 yr for the 2040s state) and the launch campaign 0.5 yr (A). Mars launches are snapped to the 26-month windows, anchored on the December 2028 window that SR-1 Freedom targets [35]; outer-planet windows recur about yearly and are rounded up. The earliest feasible launch year of a mission is then the earliest year of any state that closes it.

Table 16 gives both parts of the verdict, and Fig. 18 shows them side by side. Table 17 lists the gates of each state and Table 18 the sensitivity of the earliest launch year to the programmatic assumptions.

Mars cargo, 8 t in 450 d. The mission closes in every state: 95 kWe and 24.9 t in the 2020s, 64 kWe and 16.9 t in the 2030s and 80 kWe and 14.9 t in the 2040s (trip 428 d against the 450-d limit; Hohmann flight time 259 d). The three trajectory models agree here, and SMAD's native 2030s design is 60 kWe / 17.2 t. Its programmatic gate is the reactor and shield ground test, not the converter: three years of full-power operation are already covered by the existing Brayton evidence. The earliest launch is therefore the Aug 2039 window, with the 2030s state, the 2020s state giving the same date at a 48 % heavier stack (24.9 t). The 2040s state is lighter still but cannot fly before Jan 2046. The 100 kWe system of the original paper would be needed only for much faster trips, and the 200-d, 5 t Mars cargo of the original is not closable (Section 4.3).

Jupiter orbiter, 1.5 t in 6 yr. Under the common definition (rendezvous with Jupiter's orbit and a spiral to 30 R_J), the 2030s design is 45 kWe and 9.4 t (P(close) 1.00), the 2020s design 89 kWe and 20.4 t, which closes only with low margin because it needs a refuelled Starship with 13 % to spare, and the 2040s design 40 kWe and 7.6 t. SMAD's native results are 3–4× lighter in power (18 kWe and 6.1 t in the 2030s), because its reference ΔV counts about 1 km/s of capture (errata E-M2); with the same capture assumption, the common closure gives 29 kWe. The gate is the 70,000-h converter life test, which ends in 2037, so the earliest launch is 2041 and arrival about 2047. Figure 19 shows the 2030s trajectory: the vehicle thrusts to day 558, coasts for 512 d, thrusts again to day 1398 and makes a final 80-d coast to Jupiter's orbit at day 1478, followed by a 604-d capture spiral.

Uranus orbiter, 1 t in 10 yr. In the common planar model the transfer is direct, and the 10-yr limit is much shorter than the 16.1-yr Hohmann time. The 2020s state does not close at any power in the grid, the 2030s state closes with low margin (105 kWe, 17.9 t, P(close) 0.63, refuelled Starship only) and the 2040s

Table 16: Two-Part Verdict: Technical Closure and Programmatic Readiness by Mission and Technology State

Mission	State	P_e [kWe] / m_0 [t]	Trip [d]	P(close)	P_e 50 % (5–95)	LV margin	Technical closure	Launch vehicle	Gating item	Earliest launch
Mars cargo 8 t, 450 d	2020s	95 / 24.9	428	1.00	103 (91–133)	+29 %	Closes	SLS 1B or Starship	reactor + shield ground test (TRL 6)	Aug 2039
	2030s	64 / 16.9	428	1.00	70 (62–80)	+91 %	Closes	SLS 1B or Starship	reactor + shield ground test (TRL 6)	Aug 2039
	2040s	80 / 14.9	428	1.00	80 (62–91)	+116 %	Closes	SLS 1B or Starship	reactor + shield ground test (TRL 6)	Jan 2046
Jupiter orbiter 1.5 t, 6 yr	2020s	89 / 20.4	2082	0.51	103 (80–117)	+13 %	Closes, low margin	Starship (refuelled)	converter life test, 70,000 h	2041
	2030s	45 / 9.4	2082	1.00	54 (48–62)	+145 %	Closes	SLS 1B or Starship	converter life test, 70,000 h	2041
	2040s	40 / 7.6	2082	1.00	48 (37–54)	+205 %	Closes	SLS 1B or Starship	converter life test, 70,000 h	2047
Uranus orbiter 1 t, 10 yr	2020s	none	–	0.00	–	–	Does not close	–	converter life test, 105,000 h	–
	2030s	105 / 17.9	3469	0.63	133 (103–151)	+29 %	Closes, low margin	Starship (refuelled)	converter life test, 105,000 h	2045
	2040s	85 / 11.2	3469	0.99	103 (80–151)	+106 %	Closes	SLS 1B or Starship	converter life test, 105,000 h	2051

Technical closure from the common optimal-control closure (Monte Carlo N = 1000, seed 20251103, design re-optimised per sample): Closes if P(close) $\geq$ 0.8, low margin if $\geq$ 0.5. LV margin = best catalogue capacity at the mission C_3 over 1.15 m_0, minus one. Earliest launch = the year the state's development gates allow (Section 4.8); Mars dates are snapped to the 26-month windows anchored on December 2028. (A) development durations from the SMAD plan.

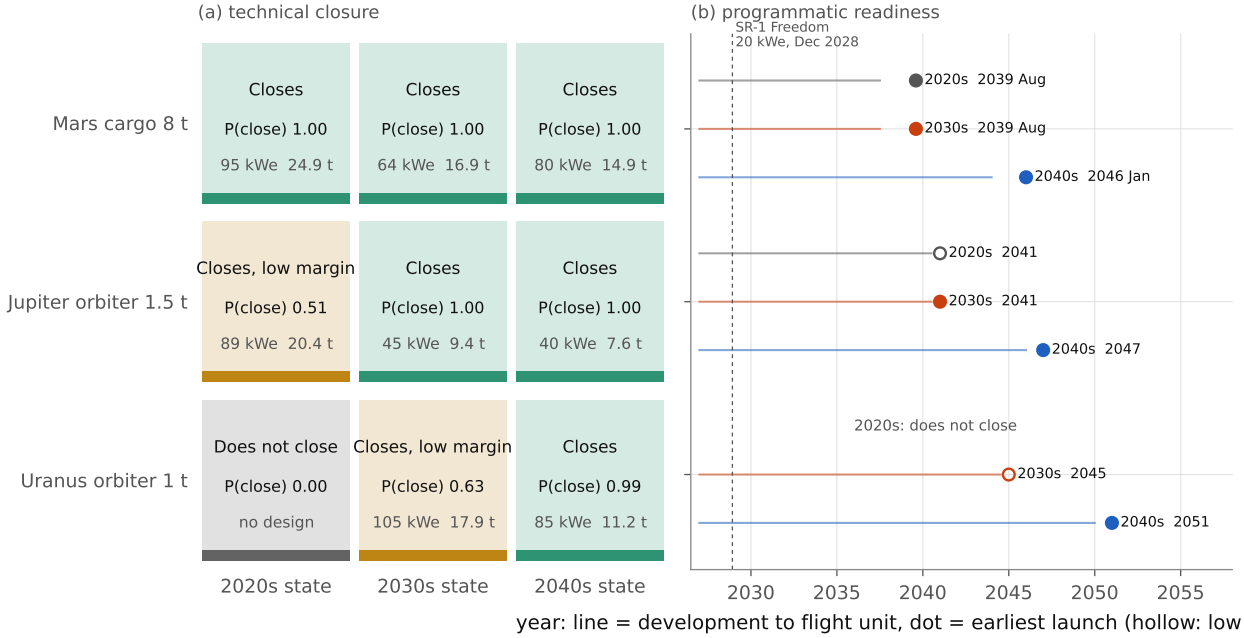


Figure 18: Two-part verdict for the three missions. (a) Technical closure by technology state, from the common optimal-control closure; the colour bar marks the closure class. (b) Programmatic readiness: development from 2027 to the flight unit (line) and earliest launch (dot; hollow for a low-margin closure). Closure never worsens with a better technology state; the later states close with more margin but fly later. Values from Tables 16 and 17.

state closes (85 kWe, 11.2 t, P(close) 0.99). The gate is the 105,000-h (12-yr) converter life test: the earliest low-margin launch is 2045 and the earliest closing launch 2051, arriving about 2060. SMAD's 17 kWe rests on a literature trajectory with gravity assists (14.2 km/s of electric ΔV) that a planar direct-transfer model cannot represent; a recent parametric study finds an optimum of 7–30 kWe for ice-giant NEP with such trajectories [10]. The Uranus rows of Table 16 are therefore a *conservative direct-transfer bound*, and the optimal-control table for Uranus is the least converged of the three.

What the two-part verdict changes. No closure class falls when the technology improves, and the reason the 2040s state is not the recommended one is now visible and correct: it closes everything, but it flies 6–7 years later. The programmatic answer is set by evidence, not by physics. For Mars the reactor ground test sets the date, and for the outer planets the converter life test does; halving the life evidence required moves the Jupiter launch to 2038 and the low-margin Uranus launch to 2039, while asking for 1.5× the mission hours moves them to 2045 and 2051 (Table 18). The launch-vehicle assumption is not binding unless refuelling slips beyond about 2035.

Trade space. To show how much room each design has, Fig. 20 plots the launch mass and trip time at closure against electric power for each technology state, with the feasible power range shaded. Mars closes over a wide band (67 to 347 kWe in the 2030s state), so its design power is a choice of mass, not a hard limit. The Jupiter band is similar (48 to 215 kWe), while the Uranus band is narrow in the 2030s state (108 to 175 kWe) and opens up only with the lighter 2040s power system (88 to 324 kWe).

Table 17: Development Gates, Flight-Unit Year and Earliest Launch of Each Technology State

Mission	State	Gates (year each is met)	Flight unit	LV gate	Launch
Mars cargo 8 t, 450 d	2020s	reactor + shield ground test (TRL 6) 2034.0	2037.5	2030	Aug 2039
	2030s	reactor + shield ground test (TRL 6) 2034.0	2037.5	2030	Aug 2039
	2040s	converter life test, 26,000 h 2036.0; reactor + shield ground test (TRL 6) 2039.0	2044.0	2030	Jan 2046
Jupiter orbiter 1.5 t, 6 yr	2020s	converter life test, 70,000 h 2037.0; reactor + shield ground test (TRL 6) 2034.0	2040.5	2030	2041
	2030s	converter life test, 70,000 h 2037.0; reactor + shield ground test (TRL 6) 2034.0	2040.5	2030	2041
	2040s	converter life test, 70,000 h 2041.0; reactor + shield ground test (TRL 6) 2039.0	2046.0	2030	2047
Uranus orbiter 1 t, 10 yr	2020s	converter life test, 105,000 h 2041.0; reactor + shield ground test (TRL 6) 2034.0	2044.5	–	2045
	2030s	converter life test, 105,000 h 2041.0; reactor + shield ground test (TRL 6) 2034.0	2044.5	2030	2045
	2040s	converter life test, 105,000 h 2045.0; reactor + shield ground test (TRL 6) 2039.0	2050.0	2030	2051

(A) Plan start 2027; critical items reach TRL 6 in 2034 (2020s and 2030s states) or 2039 (2040s state); a real-time converter life test of the mission's full-power hours starts when converter test articles exist (2029; 2033 for the 1500-K converter) unless the 50,000-h Brayton evidence already covers the mission; integration 3.5 yr (5 yr for the 2040s state); launch campaign 0.5 yr; refuelled Starship operational 2030.

Table 18: Sensitivity of the Earliest Launch Year to the Programmatic Assumptions

Case	Mars cargo	Jupiter orbiter	Uranus orbiter
baseline	Aug 2039 / Aug 2039	2041 / 2041	2051 / 2045
evidence 0.5x mission hours	Aug 2039 / Aug 2039	2038 / 2038	2045 / 2039
evidence 1.5x mission hours	Aug 2039 / Aug 2039	2045 / 2045	2057 / 2051
integration 1 yr faster	Jul 2037 / Jul 2037	2040 / 2040	2050 / 2044
reactor test 2 yr late	Oct 2041 / Oct 2041	2041 / 2041	2051 / 2045
refuelling operational 2035	Aug 2039 / Aug 2039	2041 / 2041	2051 / 2045

Each cell: earliest launch with closure P $\geq$ 0.8 / with P $\geq$ 0.5.

Solar versus fission, equal mass. To compare the power sources directly, the same 8 t Mars mission is flown in the animation data set `power_source_compare` with fission NEP, with solar-electric power (25 kg/kW, $1/r^2$) and with an RTG power system of the same mass (about 3 t). The solar craft reaches Mars orbit in 393 d against 362 d for the fission craft, and the RTG craft (7 kWe) does not get there. At Mars, solar-electric propulsion is therefore nearly competitive; the advantage of fission is distance (Fig. 2), which is the same conclusion as that of the audit.

Where the models differ from the SMAD verdicts. SMAD's native verdicts use the earlier single rule and so cannot be compared one-to-one with the two-part verdict; their technical content can. Every cell that SMAD sizes as closing also closes here, and the one cell that SMAD calls not closing, the 2020s Mars cargo (27 % closure with a 28 t stack on an assumed Starship curve), closes here because the common Monte Carlo lets the design power be re-chosen in each sample and uses the largest of three catalogue vehicles. The Mars 2040s power is also higher here than in SMAD (80 against 62 kWe at nearly equal mass): the minimum-mass optimum is flat in power, and the common optimal-control closure picks the faster end.

4.9 Comparison with flown missions and current alternatives

A verdict that NEP closes a mission says nothing about whether it is the best way to fly it. To place the results against current practice, we first list what electric propulsion and space reactors have actually demonstrated, and then compare each mission with the best alternative that exists or is studied today. Table 20 summarises the reference programmes, Table 21 the per-mission comparison and Fig. 21 the trip time against the mass delivered.

What has been demonstrated. Solar-electric propulsion is a mature deep-space technology. Dawn's ion system gave almost 11.5 km/s over about 51,250 h of operation [37], Psyche is cruising on Hall thrusters with 21 kW arrays near Earth [38] and BepiColombo completed an 8-year ion-propelled cruise in 2026 [41]. The same record also shows the failure modes that matter for a decade-long NEP mission: BepiColombo lost part of its array power and had to change route, and DART's NEXT-C was switched on once and then suspended after an electrical anomaly traced to insufficient pre-launch electrical testing [28]. Space fission power has a far thinner record. SNAP-10A ran for 43 days in 1965 and no integrated NEP system has

Table 19: Mass ledger [kg] of the minimum-mass optimal-control designs (common inputs); radiator in m^2, thruster life as fraction of the demonstrated NEXT value, times in days.

Mission	Era	P_e	m_{ps}	m_{thr}	m_{bus}	reserve	tanks	Xe	m_0	A_{rad}	N_{thr}	life	T_{helio} / T_{cap}
Mars cargo 8 t, 450 d	2020s	95	7685	747	3278	1171	366	3655	24901	94	13	0.21	363 / 65
	2030s	64	2935	511	2147	559	248	2475	16875	106	9	0.21	363 / 65
	2040s	80	2276	633	1966	487	140	1396	14898	44	11	0.14	370 / 57
Jupiter orbiter 1.5 t, 6 yr	2020s	89	7594	681	1564	984	737	7370	20429	191	12	0.78	1474 / 609
	2030s	45	2321	327	1160	381	338	3381	9408	74	6	0.77	1478 / 605
	2040s	40	1483	280	1133	290	262	2617	7564	29	5	0.68	1517 / 566
	2030s	105	4544	855	1162	656	880	8804	17902	173	16	0.79	3331 / 138
	2040s	85	2500	638	1138	428	501	5010	11214	61	11	0.64	3351 / 119

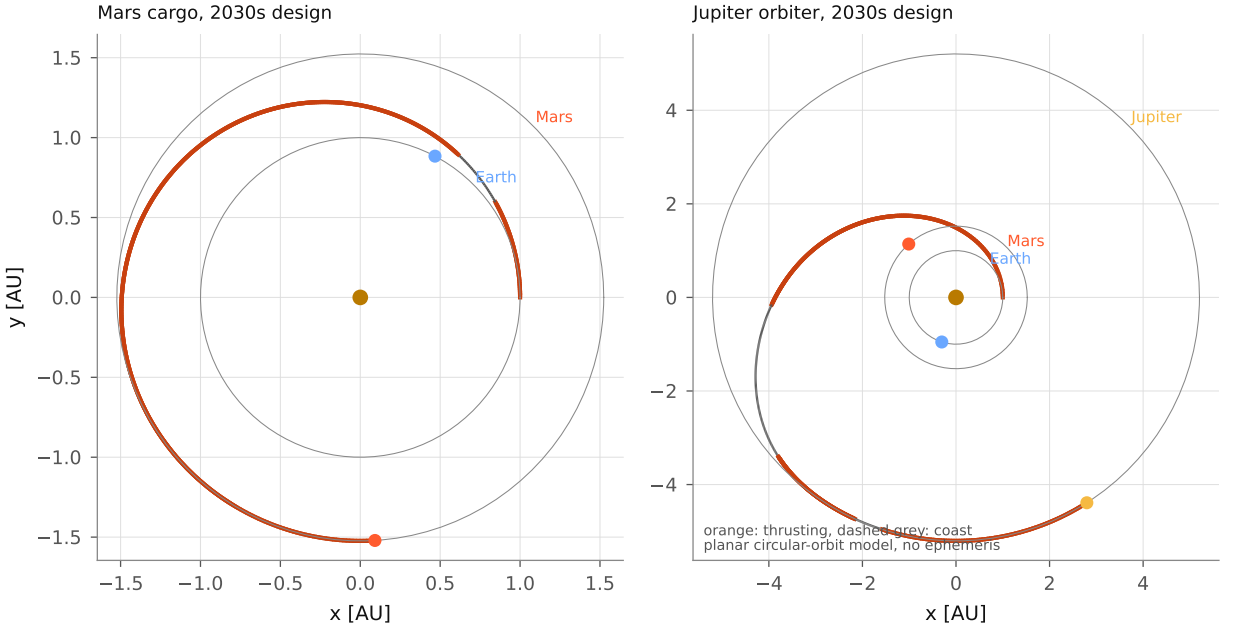


Figure 19: Optimal-control trajectories of the 2030s Mars and Jupiter designs in the planar model (orange: thrusting; dashed: coast). The capture spiral is not drawn. The Jupiter optimum coasts for a third of its heliocentric leg.

been tested since [44]; TOPAZ I flew twice at 5 kWe [45]; KRUSTY ran for 28 h at 1.5–5 kWt in 2018 [48]. JIMO, the last 100-kWe NEP programme, was deferred in 2005 over cost and complexity [47], and DRACO, a nuclear-thermal demonstrator, was cancelled in 2025, with DARPA stating that nuclear electric is "probably a more optimal long term solution" [52]. The most important change since the 2025 paper is SR-1 Freedom: a 20 kWe HALEU reactor with closed-Brayton conversion on the repurposed Gateway power and propulsion element, about 12 t, targeting launch in December 2028 and about one year to Mars [27, 35]. If it flies as planned, it would give flight evidence for exactly the reactor-Brayton-Hall chain modelled here, at one fifth of the Mars cargo power. Our readiness model does not take credit for it, because its flight unit and its mission hours are smaller than those required here; a successful SR-1 would be the strongest reason to move the 2034 TRL-6 gate earlier.

Mars cargo. For a single 8 t cargo, chemical propulsion is the better choice today. With the same launch-vehicle model and the same 20 R_M delivery orbit, one Falcon Heavy injects 11.2 t on a Hohmann transfer and places 6.8 t in orbit after a 1.35 km/s insertion burn, in 259 d (Perseverance flew a faster 203-d transfer [54]); the 2030s NEP vehicle delivers 8 t in 428 d but needs a 16.9 t stack on SLS or a refuelled Starship and could not launch before Aug 2039. NEP is therefore 1.7× slower, delivers about 17 % more per vehicle, costs an estimated \$5.7 B (4.1–7.9) against a \$0.21 B launch plus a chemical stage, and carries the risk of a TRL-4 reactor. Large solar-electric stages are the stronger electric alternative: a 150 kW SEP stage on one SLS delivers 31 t to a high Mars orbit in 4.6 yr, or 41 t in 2.8 yr with aerocapture [55], and the equal-mass comparison of Section 4.8 shows solar-electric only 8 % slower than fission at Mars. The case for NEP at Mars rests on what this comparison leaves out: a reusable tug (the reactor outlives one transfer), 64 kWe available in Mars orbit for relay or surface-support roles, and cargoes too large for one chemical launch. Refuelled Starship would change the picture again, but its Mars plans for 2026 were abandoned in February 2026 [66] and ship-to-ship refuelling has not flown [34], so no Starship Mars capability is used here.

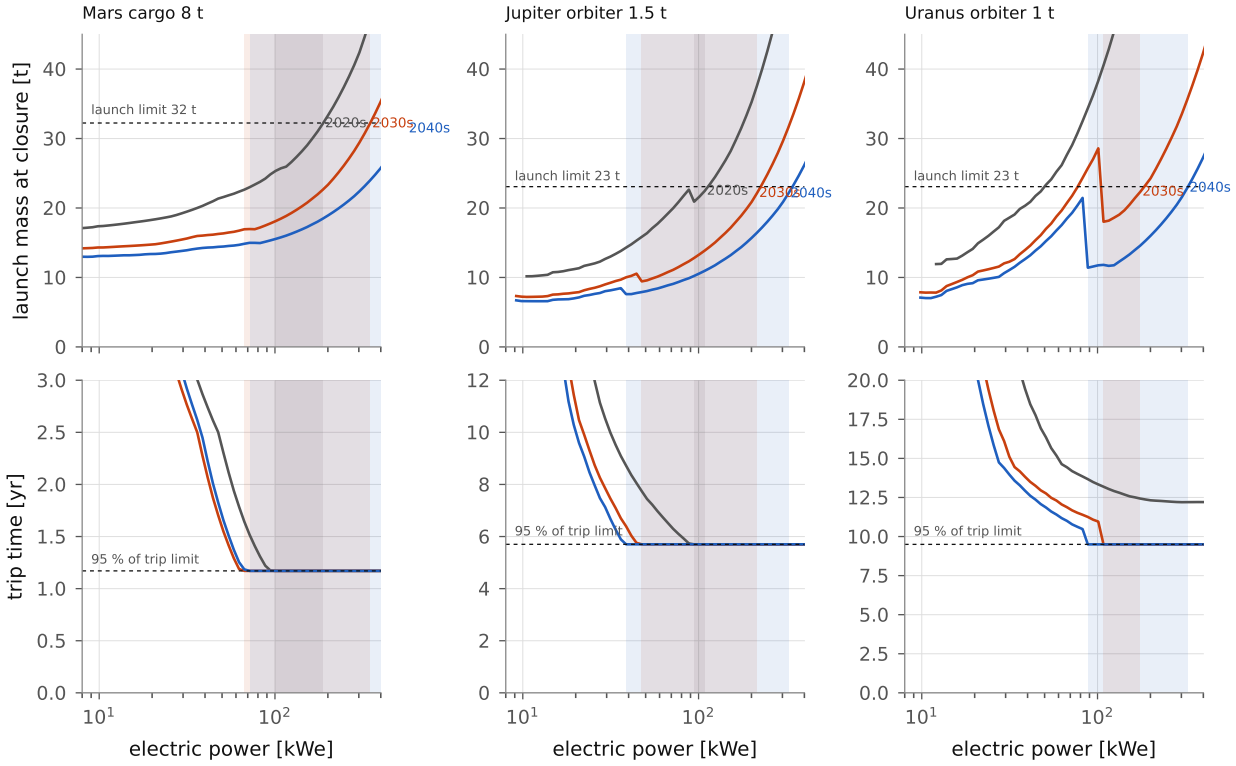


Figure 20: Launch mass (top) and trip time (bottom) at closure versus electric power for the three missions and technology states, with the feasible power range of each state shaded. Mars and Jupiter close over a wide band of power; Uranus closes only in a narrow band until the 2040s state. Common closure, optimal-control trajectories; dashed lines are the launch limit and 95 % of the trip limit.

Jupiter orbiter. NEP is not faster than the flown missions to Jupiter: Juno took 4.9 yr and Europa Clipper takes 5.5 yr with Mars and Earth gravity assists [57, 60], against 5.7 yr for the 2030s NEP design. What NEP adds is orbit and power. The NEP vehicle arrives with about 6 t of dry mass in a circular 30 R_J orbit and 45 kWe; the same circular orbit is almost unreachable chemically (a Hohmann transfer on one Falcon Heavy leaves only 0.5 t after the 4.6 km/s insertion, which is why the flown missions use elliptical orbits and moon tours), and Europa Clipper has at least 700 W at Jupiter [59], about 60 times less. At a similar estimated cost ($6.2 B, 4.5–8.6, against Clipper's $5.2 B [31]), NEP buys a science platform with an order of magnitude more data rate (Section 4.7) and the ability to change orbit at will, but not an earlier arrival: its earliest launch, 2041, is eleven years after Clipper's arrival.

Uranus orbiter. Uranus is the mission that the 2025 paper called NEP-enabling, and the comparison is the most informative. The Decadal Uranus Orbiter and Probe needed a 2031–32 Jupiter gravity assist and a 13.4-yr cruise on an expendable Falcon Heavy, at $2.8 B in the study's own estimate [23] and about $4.2 B in the independent one [63]. That window has been lost to funding, and the post-Decadal update offers a solar-electric stage with Falcon Heavy that delivers 3.0–4.2 t to Uranus orbit in 12–14 yr from 2035–2040 launches, or a refuelled Starship direct trajectory of about 10 yr [64]. With aerocapture a Starship-launched trajectory reaches Uranus in 6 yr [65], although the same update judged aerocapture too risky for the orbiter [64]. Against this, the conservative NEP design of this paper takes 9.5 yr, delivers 1 t of payload (6–9 t dry), needs 85–105 kWe and cannot launch before 2045 with low margin or 2051 with margin, i.e. it arrives around 2055–2060, about a decade after a 2035–2040 SEP mission. A study that uses gravity-assist trajectories finds that 7–30 kWe NEP would deliver the UOP payload in about 10.5 yr, or up to three times that payload [10], which shows how much our direct planar transfer overstates the power. NEP's decisive advantage at Uranus is not the trip time but the power at arrival: 85–105 kWe against about 553 W from the RTGs [64], which returns the UOP science volume in 192 h of 34-m DSN time instead of 5915 h (Section 4.7), at an estimated $6.2–8.1 B.

Animated comparisons. The animation data sets `compare_mars_alt`, `compare_jupiter_alt` and `compare_uranus_al` (site compare player) show the 2030s NEP trajectory against chemical transfers sized with the same launch-vehicle model and the same capture orbit. For Uranus they include a 10-yr direct ballistic transfer of the

Table 20: Flown and Planned Electric-Propulsion and Space-Reactor Programmes Used as Reference Points

Programme	Power	What it demonstrated / status (Nov 2026)	Source
Dawn (2007–2018), 3 NSTAR-class ion thrusters	solar, >10 kW at 1 AU	11.5 km/s, 51,250 h, 416 kg Xe	[36, 37]
Psyche (2023–), 4 Hall thrusters, one at a time	solar, 21 kW near Earth, 2.3 kW at 3 AU	240 mN; Mars assist May 2026; arrival Aug 2029	[38, 39]
BepiColombo (2018–2026), 4 T6 ion thrusters	solar, two 15-m wings	250 mN for two thrusters; cruise 8 yr against 7 planned	[40, 41]
DART NEXT-C (2021)	solar	one activation; PSE anomaly, further use suspended	[28]
Gateway PPE (paused 2026)	solar, up to 60 kW; 3 × 12 kW AEPS	bus reused for SR-1 Freedom	[42, 43]
SNAP-10A (1965)	reactor, 500 We	43 days; first reactor-powered ion thruster	[44]
TOPAZ I (1987–88)	thermionic reactor, 5 kWe	6 and 11 months in orbit	[45]
SP-100 (1983–1992)	reactor, 100 kWe goal	ground tests only	[45]
JIMO / Prometheus (2003–2005)	100 kWe Brayton, 41 kg/kWe	deferred over cost and complexity	[46, 47]
KRUSTY (2017–2018)	1.5–5 kWt, 2 Stirling at about 90 We	28 h full-power nuclear test, <$20 M	[48]
Fission Surface Power / Lunar Reactor-1	40 kWe (2022) → ≥100 kWe (2025) → ≥20 kWe (2026 draft)	three $5 M Phase 1 awards; no flight award yet	[49–51]
DRACO (2023–2025), nuclear thermal	–	$499 M agreement, cancelled 2025; DARPA: NEP “more optimal”	[30, 52]
SR-1 Freedom (announced Mar 2026)	20 kWe HALEU reactor, closed Brayton, on the PPE bus	12 t; launch Dec 2028; about 1 yr to Mars	[27, 35]
Russian TEM / Zeus	megawatt class	underfunded (2022, 2024); no verified 2025–26 milestone	[53]

kind quoted for a refuelled Starship: on the study’s assumed Starship curve it needs $C_3 = 132$ km^2/s^2 and still places 2.7 t in the 30 R_U orbit, against 0.6 t for a 16-yr Hohmann transfer on one Falcon Heavy. These cases have no gravity assists and are bounds, not mission designs.

Summary of the comparison. With current and near-term alternatives counted honestly, NEP does not shorten the trip to any of the three destinations relative to the best alternative that could launch first, and it costs at least as much. Its measurable advantages are (i) mass delivered into an orbit of free choice, (ii) tens of kilowatts at the destination, which turn into data rate and into science or relay capability, and (iii) reusability for cargo. These are real, but they are a different argument from the one the 2025 paper made.

Table 21: NEP Against the Best Current Alternatives for Each Mission

Option	Trip	Delivered [t]	Launch	Cost [$B]	Power / readiness	Available
Mars cargo, 8 t to a 20 R_M orbit						
NEP, 2030s state (computed)	428 d	8.0	16.9 t, SLS 1B or refuelled Starship	5.7 (4.1–7.9)	64 kWe at Mars; reactor and boom TRL 4 today	Aug 2039
Chemical Hohmann, one Falcon Heavy (computed, same LV model and orbit)	259 d	6.8	11.2 t injected	launch $0.21 B [23] + stage	flown technology	now
Fastest flown cruise, Perseverance [54]	203 d	–	–	–	flown	–
SEP-Chem 150 kW stage, one SLS [55]	4.6 yr	31	single SLS	–	50-kW-class SEP ready [56]; 150 kW not flown	study
SEP + aerocapture [55]	2.8 yr	41	single SLS	–	aerocapture not flown at Mars	study
Jupiter orbiter						
NEP, 2030s state (computed), 30 R_J circular orbit	5.7 yr	6.0 dry (1.5 payload)	9.4 t	6.2 (4.5–8.6)	45 kWe at Jupiter	2041
Europa Clipper, chemical + Mars and Earth assists [31, 57, 58]	5.5 yr	3.24 dry	5.7 t, Falcon Heavy expended	5.2	≥700 W solar [59]	launched 2024
Juno, chemical + Earth assist [60]	4.9 yr	1.59 dry	3.6 t	1.13	about 400 W solar	launched 2011
JUICE, chemical + assists [61]	8.2 yr	2.4 dry	6.0 t, Ariane 5	1.6 B EUR [62]	solar	launched 2023
Uranus orbiter						
NEP, 2030s state (computed, direct planar transfer)	9.5 yr	9.1 dry (1.0 payload)	17.9 t, refuelled Starship	8.1 (5.9–11.1)	105 kWe at Uranus; P(close) 0.63	2045
NEP, 2040s state (computed)	9.5 yr	6.2 dry	11.2 t	6.2 (4.5–8.5)	85 kWe; P(close) 0.99	2051
UOP: chemical + Jupiter assist, 3 RTGs [23]	13.4 yr	2.76 dry	7.2 t, Falcon Heavy expended	2.8 (indep. 4.2 [63])	553 W at arrival [64]	2031 window lost [64]
SEP stage + Falcon Heavy [64]	12–14 yr	3.0–4.2	Falcon Heavy	–	2 RTGs; SEP only to 4 AU	2035–2040 launch
Refuelled Starship, direct [64]	about 10 yr	> 4	many tanker launches	–	refuelling not yet demonstrated [34]	–
Starship + aerocapture [65]	6 yr	–	–	–	aerocapture judged too risky for UOP [64]	2042 trajectory
NEP 10–40 kWe, gravity-assist trajectories [10]	about 10.5 yr	0.25–1.25 payload	–	–	optimum 7–30 kWe	–

Costs as published (year of dollars differs; Juno and Clipper not stated); NEP costs are this paper's analogy model with a nuclear adder, P50 (P10–P90), Section 4.7.5. Chemical delivered mass for the Jupiter and Uranus missions is the flown or studied dry mass in their own (elliptical) orbits, not the NEP capture orbit.

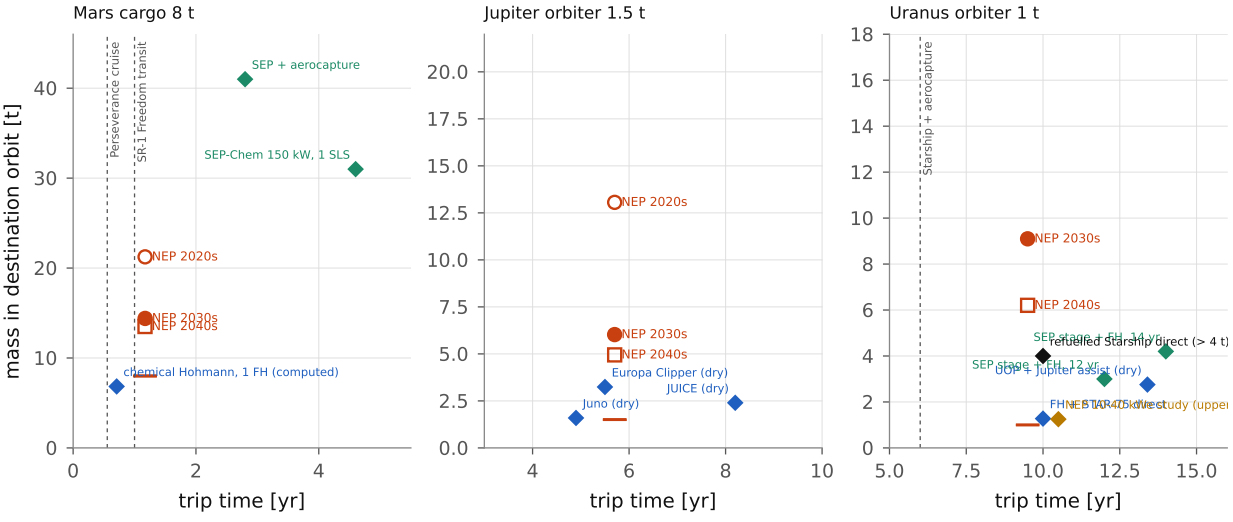


Figure 21: Trip time versus mass delivered to the destination orbit for NEP (orange; dot = dry vehicle, bar = payload, by technology state) and for the current alternatives (blue: chemical with or without gravity assists; green: solar-electric; black: refuelled Starship; dashed lines: trip times without a published mass). For Mars and Jupiter, NEP is not faster than the flown chemical missions; its gains are delivered mass into a chosen orbit and power at the destination. Sources in Table 21.

4.10 Novel methods: what holds

Design ideas were collected from three places: the simulation's novel-methods case (case E), the option screen of the SMAD study (its section 7a, a paired Monte Carlo of 17 uncertain inputs with $N = 177$ and the keep rule of a median gain $> 2\,\%$ and $P(\text{lighter}) \geq 0.8$), and the reconcile runs of this paper. Each idea was then checked against the literature with a strict rule: a source counts only if its text or abstract page was opened and the statement seen (Appendix A).

Kept. Table 22 lists the options that pass the screen. Three points limit how far they can be trusted. (i) Every gain is a *mass* gain on a power system that is only 20–45 % of the dry mass; the gains are not additive, and none removes a technology gate of Section 4.6. (ii) Two of the largest (the liquid-droplet radiator and the 1500 K Brayton) are TRL 3 research options with no flight or long-duration data [14, 67]. The options that are both mass-positive and at TRL ≥ 4 are the carbon-fibre radiator, radiation-hardened avionics, the xenon-tank gamma credit, PPU-heat reuse and a battery for peak loads. (iii) Most are not new. The literature check found precedent for the xenon-tank shield (mercury propellant in a NEP tug shield [68], and xenon and tanks credited in a Mars NEP shield [14]), the droplet radiator [69], the high-temperature Brayton [14] and the carbon-carbon radiator goal [15]. What this paper adds is the quantification on one consistent mission set, not the ideas themselves.

Simulation case E (quantified against the Mars and Jupiter designs).

- *Free specific-impulse optimum* (precedent: the Stuhlinger velocity and optimum-I_{sp} results in [3], and analytic optima for ion, MPD and arcjet vehicles in an abstract-only source [70]). For the Mars cargo case at a 364-d trip, the free-I_{sp} optimum (2400 s, 141 kWe) delivers 5800 kg against 5668 kg for HERMeS (2802 s) and 5537 kg for NEXT (4170 s), a gain of +2.3 % and +4.7 % respectively. At 430–500 d the optimum moves to 3600 s, and the gain over NEXT falls to 1.5–2.4 %. Figure 7 shows that, as the power system gets lighter, the Mars optimum moves from about 2400 s (2020s) to 4050 s (2040s), and for the 2040s Jupiter case it moves beyond the 4300 s limit of the efficiency fit, where the number is an extrapolation. **Verdict: a known rule, kept as a design rule; the gain is small.**
- *Jettisoned propellant tanks* (staged tanks; [71], abstract only, reports about 10–13 % lower required specific mass for a power-limited vehicle at a payload ratio of 0.1). Halving the tank mass fraction lowers the initial mass by 0.9 % for Mars and 4.2 % for Jupiter (paired band −4.5 to −4.0 %). **Verdict: supported for the long, propellant-heavy Jupiter case and negligible for Mars; it costs a separation mechanism, and precedent exists.**
- *Jupiter gravity assist for Saturn and Uranus* (patched-conic upper bound). The electric ΔV falls from 15.7 to 10.3 km/s for Saturn (5.4 km/s saved) and by 4.8 km/s for Uranus, while the flight time roughly doubles (2208 to 4651 d for Saturn); the phasing (Jupiter–Saturn synodic period 19.9 yr) is not modelled. The literature check found no published electric-ΔV saving of this kind for NEP. It found chemical Jupiter-assist routes to Uranus of about 13 yr that are available only in certain launch windows [72, 73], and published NEP trajectories that use an Earth assist instead [74]. **Verdict: reported as a bound only, not claimed.**
- *Multiple reactors.* Using 2, 3 and 4 units raises the initial mass by 9.7, 16.7 and 22.4 % (Mars) for single-fault tolerance, and losing one of two reactors doubles the Mars trip (555 to 1070 d). Precedent: [75] (about 2–4.5 kg/kWe at MWe scale). **Verdict: a cost, not a gain.**

Dropped. The options in Table 23 and the following results failed, and they are kept as negative results: hybrid chemical capture at Jupiter (initial mass +25 %, and +23 to +29 % with a chemical I_{sp} of 300–330 s); variable I_{sp} (an extra final mass at equal burn time of 3 kg at Mars and 110 kg at Jupiter, an upper bound, since the optimum constant I_{sp} is already near 4200 s); coasting instead of spiralling (0.1–2.2 % propellant saved for 11–123 % more time); perigee-only thrusting for Earth escape (not escaped after 1500 d at 10^{-3} m/s^2); thruster-cluster sizing beyond the power limit (0 %); lithium AF-MPD and hydrogen arcjet propellants (heavier or unclosable); and shielding only an electronics vault (no effect, because the half-angle is set by the radiator wings).

Reconcile-run results. (a) With the optimal-control solver, the propellant needed at a fixed trip time equals the burn-coast-burn value to within -0.9–0.7 % (Table 7). The loss of the family is in *time*, not propellant, because the impulsive plateau is reached; the minimum time is 3–46 % shorter (median 30 %) at the same thrust. (b) In the tier development plan, the converter life test is the critical path in both tiers. Accepting 50,000 h instead of 100,000 h moves the 2030s flight unit from 2044 to 2038, at the price

Table 22: Supported design options from the SMAD study's paired Monte Carlo (near-term tier, N = 177, fixed seed; bands are lower bounds on the true spread) with the literature check. Gains are not additive. None removes a technology gate.

Idea	Change in launch mass, %: median [p10, p90] (i Mars, ii Jupiter, iii Uranus)	Cost / risk	Literature precedent (strict check)	Precedent	Verdict
Liquid-droplet radiator (LDR)	I -15.6 [-29.0, -8.0]; II -5.0 [-12.0, -0.8]; III -8.1 [-20.0, -1.8]	TRL 3; fluid loss, stream stability	Mattick & Hertzberg 1981 [69]; ground tests, not flown [67]	PRECEDENT (ground tests)	Keep
Higher-temperature Brayton (1500 K turbine inlet) with smaller radiator	I -11.6 [-26.5, -5.0]; II -2.5 [-6.5, -0.5]; III -4.0 [-11.8, -0.9]	TRL 3; new fuel form and alloys	Mason et al. 2021 [14]: 2.9 vs 1.6 MWe in the same radiator, "considerable development risk"	PRECEDENT	Keep
Deployable carbon-fibre heat-pipe fin radiator	I -14.7 [-26.2, -8.1]; II -7.5 [-14.2, -3.9]; III -11.1 [-21.8, -5.6]	TRL 4; loop and armour mass not demonstrated at 2.2 kg/m^2	NASA C-C radiator goal about 2 kg/m^2, no measured value [15]	PRECEDENT (goal)	Keep
Radiation-hardened payload/avionics (limits x4: 100 krad, 4e11 n/cm2)	I -4.6 [-9.0, -2.5]; II -3.0 [-5.0, -1.7]; III -4.1 [-8.3, -2.2]	neutron damage at 4e11 n/cm^2 limits optics and power parts	SMAD Table 8-7 [5]; no NEP-specific study opened	PARTIAL	Keep
Use the xenon tank as a gamma shield (tank between reactor and payload)	I -10.1 [-21.2, -5.3]; II -4.0 [-6.7, -2.3]; III -8.8 [-17.3, -4.7]	tank on axis in the umbra; credit shrinks as xenon is used	mercury propellant in the shadow shield of a NEP tug [68]; Xe and tanks credited qualitatively in [14]	PRECEDENT	Keep
Use PPU waste heat for bus heating (heaters off)	II -1.4 [-1.9, -1.2]; III -2.5 [-4.7, -2.0]	plumbing; heat only while thrusting	loop heat pipes cut NEXT PPU survival-heater power 183 to 35 W, not the same idea	PARTIAL	Keep (small)
Battery to shave payload power peaks	II -7.1 [-9.8, -6.0]; III -6.0 [-9.7, -5.0]	45 kg battery, cycling over 10 yr	standard practice for variable loads; no NEP source opened	NONE FOUND (routine)	Keep

Table 23: Options that failed the test (kept as negative results).

Idea	Change in launch mass, %	Result
Shield only an electronics vault (payload-zone radius 1.5 -> 0.6 m)	I +0.0 [+0.0, +0.0]; II +0.0 [+0.0, +0.0]; III +0.0 [+0.0, +0.0]	DROP
Lithium propellant at 7,000 s (AF-MPD) instead of xenon	I +144.0 [+72.1, +258.2]; II +18.6 [+3.6, +61.0]; III +12.4 [-7.1, +58.2]	DROP
Hydrogen arcjet at 1,000 s	I +77.7 [+44.3, +139.8]	DROP; DROP (does not close)

of a 1.4–2.1× life gap on the outer-planet missions. This is a plan derived from assumed durations, not a forecast.

4.11 Sensitivity and "what would need to be true"

Single-assumption variants (this paper). Table 24 re-solves the 2030s minimum-mass designs with one assumption changed at a time. Halving the tank mass fraction saves 1 % of the Mars mass, about 5 % for Jupiter and 12 % for Uranus, so the propellant-heavy cases gain most. A 20 % lighter power system saves 6 % (Mars), 13 % (Jupiter) and 24 % (Uranus) of the initial mass. The capture definition is the strongest lever for Jupiter, where the power drops from 45 to 29 kWe, and it matters little for Uranus, where the heliocentric transit time dominates. The "minimum-mass" design is a flat optimum: a lighter power system pushes the best power *up* for Mars (71 against 64 kWe), because a faster trajectory then costs less mass.

Tornado of the minimum launch mass. Figure 22 extends these variants to a symmetric one-at-a-time sensitivity on the common optimal-control closure, re-optimising the design power for each change. For Mars the payload (±14 %) dominates, followed by the trip-time limit and the power-system specific mass (±6 %), because the 8 t cargo is half of the stack. For Jupiter the power-system mass leads (−13/+14 %). For Uranus a 20 % heavier power system or a 10 % shorter trip limit removes the closure altogether, and a 20 % lighter power system saves 24 %; the Uranus design sits close to the edge of its narrow feasible band (Fig. 20).

What the verdicts hinge on (SMAD study, section 9.5). Table 25 lists the assumptions whose removal alone flips a verdict. In the 2030s era the verdicts depend on *evidence* (a 100,000-h converter test rather than 50,000 h, and critical items at TRL 6 rather than 4) and on *launch availability* (a Falcon-Heavy-

Table 24: Single-assumption variants of the minimum-mass design. The capture definition is the largest lever for the outer planets.

Mission	Variant (optimal-control closure, 2030s)	P_e [kWe]	m_0 [t]	Δm_0 [%]
Mars cargo 8 t, 450 d	baseline (2030s, optimal control)	64	16.9	-
	tank mass fraction halved (staged tanks)	64	16.7	-1.1
	power-system mass −20 %	71	15.8	-6.2
Jupiter orbiter 1.5 t, 6 yr	baseline (2030s, optimal control)	45	9.4	-
	tank mass fraction halved (staged tanks)	43	9.0	-4.8
	power-system mass −20 %	43	8.2	-12.8
	capture ΔV 1 km/s (SMAD-like) instead of the spiral	29	6.6	-29.6
Uranus orbiter 1 t, 10 yr	baseline (2030s, optimal control)	105	17.9	-
	tank mass fraction halved (staged tanks)	94	15.8	-11.8
	power-system mass −20 %	89	13.5	-24.4
	capture ΔV 1 km/s (SMAD-like) instead of the spiral	92	14.9	-16.7

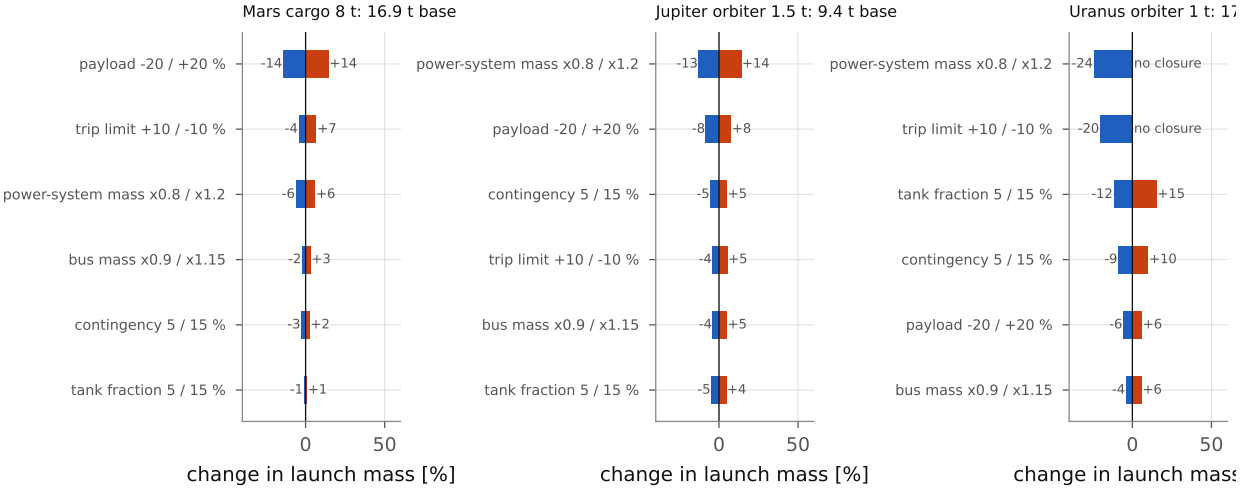


Figure 22: One-at-a-time sensitivity of the minimum launch mass of the 2030s designs to six inputs (blue: low value, orange: high value). Payload drives Mars, power-system mass drives Jupiter, and Uranus loses closure with a heavier power system or a shorter trip. Common optimal-control closure, design power re-optimised; ranges from the Monte Carlo and the SMAD register (v3/tornado.py).

only world loses the Mars and Jupiter closures). The mass assumptions change the launch mass by 13–58 %, but they do not flip a verdict by themselves except where the stack nears a vehicle limit. In the 2040s no single technology assumption flips a verdict (each costs 0–7 % of launch mass), and the only mission-enabling dependency is the refuelled-Starship capacity for the 20 t Mars cargo. These are the SMAD study's single-rule verdicts; under the two-part framework of Section 4.8 the evidence items (A1, A2) move the earliest launch year rather than the closure class (Table 18).

What would need to be true, in one list. (1) A flight-like reactor and shield ground test at power, and a deployed full-scale radiator (the 2030s milestones). (2) A converter life demonstration of at least the mission's full-power life (3, 8 and 12 yr for Mars, Jupiter and Uranus, i.e. 26,000, 70,000 and 105,000 h). (3) A launch vehicle for 15–25 t at $C_3 \approx 8$ (Mars) or 6–10 t at $C_3 = 40$ (Jupiter). (4) For the outer planets, a capture orbit and a transfer of the kind that the planar direct model of this paper over-penalises (gravity assists, lower capture orbits) or, failing that, power of order 100 kWe for Uranus. None of these is a physics limit; all are engineering and programmatic.

Uncertainty bands. The closure probabilities of Table 16 are 1.00 for most designs, so the informative spread is the sampled design power (5th–95th percentile in the table), which brackets the nominal value by roughly ±20–40 %. The fixed-hardware alternative (Table 26) is of order one half by construction. Neither includes structural model error (shield, Carnot fraction, bus scaling), which is larger.

5 Discussion

The results of Section 4 show that NEP closes the three missions, that when it could fly them is set by test evidence rather than by mass closure, and that against current alternatives its advantage is power and orbit

Table 25: SMAD study, section 9.5: the assumptions whose removal flips a verdict (all other single removals leave the verdict unchanged).

Era	Mission	ID	If this is NOT true	Base	Variant
2030s	(i) Mars cargo 8 t	A2	critical items stay at TRL 4 (no TRL-6 milestones)	Possible	Marginal
2030s	(i) Mars cargo 8 t	A3	power-system mass is the FSP-derived 142 kg/kWe curve, not the scaled reference	Possible	Marginal
2030s	(i) Mars cargo 8 t	A5	only Falcon Heavy available (no SLS, no Starship)	Possible	Does Not Close
2030s	(ii) Jupiter orbiter 1.5 t	A1	converter evidence only 50,000 h (not 100,000 h)	Possible	Marginal
2030s	(ii) Jupiter orbiter 1.5 t	A2	critical items stay at TRL 4 (no TRL-6 milestones)	Possible	Marginal
2030s	(ii) Jupiter orbiter 1.5 t	A5	only Falcon Heavy available (no SLS, no Starship)	Possible	Not Possible
2030s	(iii) Uranus orbiter 1 t	A1	converter evidence only 50,000 h (not 100,000 h)	Marginal	Not Possible
2030s	(iii) Uranus orbiter 1 t	A5	only Falcon Heavy available (no SLS, no Starship)	Marginal	Not Possible
2040s	(vi) Mars heavy cargo 20 t	B7	Starship 100 t instead of 150 t (no refuelling gain)	Marginal	Does Not Close

Table 26: Alternative closure probability with the minimum-mass hardware frozen (power, thrusters, m_0) and inputs sampled: by construction about one half, because the design sits on the mass boundary; the columns show which constraint fails. The paper's verdicts use the re-optimised probability of Table 16.

Mission	Era	P(close), hardware fixed	fail: mass	fail: time	fail: life	fail: launch
Mars cargo 8 t, 450 d	2020s	0.21	0.55	0.52	0.00	0.00
	2030s	0.21	0.53	0.53	0.00	0.00
	2040s	0.26	0.74	0.00	0.00	0.00
Jupiter orbiter 1.5 t, 6 yr	2020s	0.21	0.64	0.04	0.39	0.00
	2030s	0.21	0.66	0.03	0.37	0.00
	2040s	0.19	0.73	0.00	0.28	0.00
	2030s	0.18	0.69	0.00	0.39	0.00
	2040s	0.18	0.75	0.00	0.27	0.00

at the destination rather than trip time. This section examines how far those findings can be trusted. It first lists the objections that a hostile reviewer would raise and then the numerical limits that remain.

5.1 What a reviewer would challenge

A hostile reading of this paper raises the following eighteen objections. Each was checked against the numbers, and the status says whether it was fixed, bounded or remains open.

1. **"The optimal-control trajectories are not proven optimal."** Open, bounded. A direct single-shooting search with 45 starts and a 20-node refinement returns feasible trajectories, and hence *upper* bounds on the minimum time and propellant. Re-integration with an independent integrator and convergence in the number of nodes were checked (Section 3), but no indirect-method (costate) certificate was computed. The real optimum can only be faster.

2. **"Planar, circular, coplanar planets with free phase ignore ephemerides, inclination and window waiting."** Open. The trip time counts from Earth escape to the capture orbit. The wait for a window (Mars synodic period 780 d), Earth escape from low orbit (7.7 km/s of ΔV if spiralled, excluded here by the stated C_3 launch) and the inclination (1.3–1.9°) are not costed. They add schedule and some ΔV, and a Lambert-based ephemeris model is the necessary next step.

3. **"The Jupiter verdict depends on how capture is defined."** Confirmed and quantified. With rendezvous plus a spiral to 30 planetary radii, the 2030s Jupiter orbiter needs 45 kWe; with a 1 km/s capture (the SMAD figure), the same closure gives 29 kWe. The truth lies between the two and depends on the science orbit and on whether an Oberth-assisted capture is feasible at low thrust, which it is not at these accelerations (E6 of the simulation: a single periapsis pass delivers metres per second).

4. **"You only unified the trajectory model, not the hardware."** True. The power-system mass tables are those of the SMAD study, which is conceptual: the shield model is conservative by a factor of 1.9 on mass against JIMO, the Brayton Carnot fraction is held constant with temperature, and the specific-mass fit is not valid above about 300 kWe. The agreement of the common closure with the SMAD native results (Table 8) shows consistency of method, not correctness of the hardware numbers. The only independent hardware check is the published NASA 20 kWe mass-equipment list [10].

5. **"The technology tiers are speculation."** The 2020s state is what exists or is funded, while the 2030s and 2040s states are conditional on milestones (Table 25, Fig. 10) and are labelled as such. Three of the four 2040s ingredients (1500 K conversion, 2.2 kg/m^2 radiators, 30,000-h high-I_{sp} thrusters) have no demonstration in the sources read.

6. **"Verdict thresholds are arbitrary, and a better technology once scored worse."** Fixed and bounded. The earlier single rule capped the 2040s designs at "marginal" through their TRL-5 items, so a lighter design scored below a heavier one. The two-part verdict reports closure (thresholds 0.8 and 0.5, our judgement) and readiness (a year) separately, enforces monotonic closure by dominance and gives the underlying numbers (Table 16), so a reader can apply another rule.

7. **"The launch years rest on assumed durations."** True. The reactor ground test, converter test start, integration and refuelling dates are taken from the SMAD plan or assumed (A). Table 18 shows that the outer-planet years move by -3 to $+6$ yr across plausible evidence requirements; the Mars year moves only with the reactor test and integration. None of the dates is a forecast, and none accounts for funding.

8. **"Closure probability is 1.00 almost everywhere, so it is uninformative."** Correct for the re-optimised definition: the uncertain inputs move the design power by tens of percent, but the launch capacity is not binding. The fixed-hardware probability (Table 26) is about one half by construction, because the design sits on the mass boundary. The informative quantity is the sampled power in Table 16.

9. **"Launch capacities are partly assumed."** The Falcon Heavy curve is a fit to one published point, SLS Block 1B availability is uncertain and the refuelled Starship curve is a placeholder, and the verdicts for the heavy Mars cases depend on the last two. Falcon-only availability flips the 2030s Mars and Jupiter verdicts (Table 25, rows A5).

10. **"Thrusters are modelled at constant I_{sp} and efficiency."** True. There are no throttle curves, and the NEXT hardware operated at 3270 s for Mars is taken from the SMAD study. The generic efficiency fit is used only between 1800 and 4300 s.

11. **"Power decay is ignored in the tables."** The optimal-control tables use constant power, but the closure applies the mean degradation factor to the thrust power (3.4×10^{-3}/yr, up to three times that in the Monte Carlo), which is a 1–3 % effect over 6 years.

12. **"There is no flight validation."** No NEP vehicle has flown. The nearest benchmark, Dawn, agrees to -7.5 % in xenon and -22 % in thrust hours (Section 3).

13. **"Bus mass is frozen at the SMAD design point."** True (errata E-S3). A $\times0.9$–1.15 multiplier is sampled, which brackets a 15 % error.

14. **"The 200-day Mars result depends on the paper's alpha convention."** It does not depend on the power-system model. The minimum time of 200 d needs an initial acceleration of 6.3e-04 m/s^2 and a propellant fraction above one half, so no mass model with $\alpha \gtrsim 10$ kg/kW closes. The lower bound on α for closure is given in `results/oc_gap.json`.

15. **"The novelty gains come from a small paired Monte Carlo and mostly lack primary sources."** Yes: $N = 177$, the bands are lower bounds and several precedents are abstract-only (Table 22). The table lists only what passed and flags the rest.

16. **"Cost, politics and nuclear approval are absent."** Partly fixed. Cost is now an explicit analogy model with a nuclear adder and a stated uncertainty (Section 4.7.5), but its data set is five missions in mixed-year dollars and the power-system adder is an assumption, so it is a cost *class*, not an estimate to budget against. Launch authorisation remains a schedule item. A "closes" verdict means technically closable, not affordable or approved.

17. **"The reliability numbers look alarmingly low."** They are lower bounds on what can be *demonstrated* from the test evidence at 60 % confidence, not predictions of how often the hardware fails (Section 4.7). They show how much more testing a decade-long mission needs, which is the point.

18. **"The comparison with alternatives mixes metrics."** Partly true. Flown chemical missions report dry mass in their own elliptical orbits, the post-Decadal Uranus study reports delivered mass, and the NEP results are delivered into circular capture orbits. Table 21 states each basis, and the computed chemical cases use the same launch model and orbit as NEP.

5.2 Sources of error and model limitations

Numerical limits that remain. The tabulated propellant fractions carry solver noise of a few percent in the lowest-acceleration rows, where some nodes did not converge and were filled from neighbours (nodes filled per table: Mars isp3270 13, Mars isp4170 16, Jupiter 12, Uranus 43); designs near those rows are less certain than the tabulated values suggest. A direct optimal-control solve at the design points differs from the table by at most 9 % in propellant fraction (the table is the more conservative). Scaling the tabulated propellant fraction by ±5 % moves the minimum-mass design by under 1 % (Mars), about 2.5 % (Jupiter) and −11/+14 % (Uranus) in mass (`reconcile/selfcheck.py`). The Uranus rows are therefore the least certain numbers of the paper, and the minimum-time table for Uranus is bounded by the solver's time limit.

What was and was not re-run from scratch. The final pass of `run_all.py` rebuilt three of the five optimal-control tables from scratch (their cache keys had changed with a code revision) and ran every later stage (validation, gap study, reconciliation and Monte Carlo, animation data, figures, tables, this PDF) in sequence; `determinism_hashes.json` records its digests. Determinism evidence: (i) the Jupiter table and the Mars $C_3 = 0$ (2000 s) table rebuilt from scratch are bit-identical to their earlier builds (every minimum time and propellant fraction equal); (ii) the burn-coast-burn gap study (`oc_gap.json`, 22 direct optimal-control solves) re-ran to a byte-identical file; (iii) two re-solved cells of the Mars 4170 s table equal the stored values exactly (`reconcile/results/determinism_check.json`). The Mars 3270 s table (2020s and 2030s Mars designs) had first been built with an earlier start set (45 steering starts instead of the current 18); the rebuilt table differs from it by up to 3.5 % in minimum time (mean −0.4 %), and its least-converged propellant nodes differ by much more, which is the solver noise discussed above. The design points are not affected: the full reconciliation output (all designs, Monte Carlo distributions and variants) is identical to the previous run except two Mars escape masses that differ in the ninth significant digit. All numbers in this paper use the rebuilt tables. The Mars 4170 s and Uranus tables were not rebuilt in the final pass.

6 Conclusion

This work audited, reproduced and rebuilt the 2025 analysis of nuclear electric propulsion, reconciling two independent models on one documented input set and checking their trajectories against a new optimal-control solver.

At the level of the original study, the headline numbers do not follow from its own equations: Table 1 is off by 0.27 for NEP and by 0.53 for the other rows, the constant-mass propellant exceeds the spacecraft mass, the asymptotes of Fig. 8 lie at 112 and 130 d rather than 150 and 250 d, and the ten-RTG mission can never complete. Once corrected, NEP is about 10× faster than solar at the same thruster power, and it is decisively better only beyond 2–3 AU at equal power-system mass. At Mars, a solar-electric craft of equal power-system mass is only about 8 % slower (393 against 362 d).

At the level of the models, the simulation and the SMAD study are reconcilable for Mars, to 5–15 % in power and mass, once the simulation's failed trajectory-table nodes (35 of 63 for Mars at $C_3 = 8.2$) are removed. Its "closes in every tier" result was an artefact, and its Mars 200-day result is withdrawn. The two models are *not* reconcilable for Jupiter and Uranus without choosing a capture definition and a transfer type, which differ between them by a factor of 2–6 in power. The burn-coast-burn family is a safe upper bound: the optimal-control solver finds the same propellant at equal time and trip times 3–46 % shorter (median 30 %), with the gap growing with thrust acceleration. This bound is not proven optimal.

At the mission level, on common inputs, a Mars cargo of 8 t in 450 d needs about 95, 64 and 80 kWe in the 2020s, 2030s and 2040s states (it is power-limited, not mass-limited), and a 1.5 t Jupiter orbiter needs 89, 45 and 40 kWe. The verdict is now reported in two parts, technical closure and programmatic readiness, so that a better technology state can no longer score worse. Mars closes in every state, Jupiter closes from the 2030s state (low margin in the 2020s) and Uranus closes with low margin in the 2030s state and fully in the 2040s state. The earliest feasible launches are Aug 2039 for Mars, gated by the reactor and shield ground test, 2041 for Jupiter, gated by a 70,000-h converter life test, and 2045 (low margin) or 2051 for Uranus, gated by a 105,000-h test. Of the design ideas, the supported ones are the xenon-tank gamma credit, the carbon-fibre radiator, radiation-hardened avionics, PPU-heat reuse and tank staging for long missions, while the liquid-droplet radiator and the 1500 K Brayton are TRL 3 research options.

Against flown missions and current alternatives, NEP is not the fastest option to any of the three destinations. One Falcon Heavy places 6.8 t in the same Mars orbit in 259 d, Europa Clipper reaches Jupiter in 5.5 yr with about 700 W, and a solar-electric Uranus mission launched in 2035–2040 would arrive about a decade before the earliest NEP orbiter. What NEP adds is mass delivered into an orbit of free choice and tens of kilowatts at the destination: a 2 kW Ka-band link returns the Uranus Orbiter and Probe science

volume in 192 h of DSN time instead of 5915 h. The operations-level analyses show that the evidence, not the hardware, limits the reliability that can be claimed: demonstrating a converter reliability of 0.9 needs about 8.7 times the mission hours of failure-free testing. Every design is a flagship-class mission, at an estimated \$5.7, \$6.2 and \$8.1 billion (P50) for the 2030s Mars, Jupiter and Uranus designs.

In general, the results demonstrate that the advantage of NEP is real but comes from distance and from power at the destination rather than from the nominal power factor or the trip time, and that when a mission can fly is decided by test evidence rather than by mass closure. The original study's claims are therefore not supported in the form or by the margins in which they were stated. The main open uncertainties are the capture and transfer definition for the outer planets, the development durations behind the launch years, the cost model, the conceptual hardware masses (shield, conversion), the launch-vehicle curves for the heavy cases, and the unproven global optimality of the trajectories. Future work should address these uncertainties, beginning with the Lambert-based ephemeris model identified in Section 5 and gravity-assist trajectories for Uranus, and should update the readiness gates once SR-1 Freedom has flown.

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A Reproducibility Statement

Code and data. All inputs are kept in `NEP_rebuild/data` (parameters with source and status), `sim` (trajectory simulation), `smad` (SMAD design study) and `audit`, and this paper adds only files under `NEP_rebuild/paper`. The command `python paper/run_all.py` (plain `python`; numpy, scipy, matplotlib, numba; xelatex) regenerates the optimal-control tables (`reconcile/build_tables.py`), the solver validation, the burn-coast-burn gap study, the reconciliation with its Monte Carlo (seed 20251103, Latin hypercube), the animation data, all figures in the paper and the dark-theme sets, all tables and macros, and this PDF. No number in the text is typed by hand. With `--clean`, the tables are rebuilt from scratch.

Determinism. The solver has no random element (fixed start list, SLSQP, fixed-step RK4), and the only use of random numbers is the seeded Latin hypercube. The file `paper/determinism_hashes.json` records SHA-256 digests of the results files (and of their numerical content rounded to nine significant digits), of every optimal-control table array and of the inputs, and a clean rebuild must reproduce them (Section 5 states what was and was not re-run).

Original packages untouched. The `sim` and `smad` packages are imported read-only, and the sim trajectory cache is read from a copy in `reconcile/cache_bcb`. All defects found in them are documented in `reconcile/errata.md` and corrected only inside the reconcile code.

Sources actually read. Only sources whose full text, abstract page or specific tables were opened are cited, and the reference list marks abstract-only items. The following were not opened and are therefore not cited as evidence: the original Irving and Blum (1959) report, Goebel and Katz, Edelbaum (1961), Kluever, Betts and Conway on low-thrust optimisation, and the Purdue work on NEP gravity assists (access was blocked). Standard results attributed to them (the Stuhlinger optimum and the tangential-spiral $\Delta V = |v_{c1} - v_{c2}|$) are instead checked numerically in Section 3 or sourced to the lecture notes and abstract pages listed. Values recalled without a document (SNAP-10A, Europa Clipper mass, xenon production) are marked as such in the SMAD study and are not load-bearing for any verdict here.

Statuses used throughout. *Computed* denotes the output of one of the models, *literature* a value read in a cited document, and *assumed* a stated assumption with a range. The technology tiers (2020s, 2030s, 2040s) are technology *states* with stated milestones, not forecasts of dates.

References

[1] Miguel Villax. Mission-enabling performance analysis of nuclear electric propulsion for deep space exploration. Course paper, SPC 536, University of Michigan, 2025. The paper audited and rebuilt here.

[2] NASA. Multi-mission radioisotope thermoelectric generator (MMRTG) fact sheet. NF-2020-05-619-HQ, 2020. URL https://science.nasa.gov/wp-content/uploads/2024/02/mmrtg-factsheet-updated-5-18-20-1.pdf.

[3] P. Lozano. Session 3-4: Mission analysis for electric propulsion. In *MIT OpenCourseWare 16.522 Space Propulsion, lecture notes*, 2015. URL https://ocw.mit.edu/courses/16-522-space-propulsion-spring-2015/0a041f5862654c42025184fe2abe2cc8_MIT16_522S15_Lecture3-4.pdf. Opened: thrust-power relation and the Stuhlinger velocity.

[4] J. S. Sovey et al. Development of an ion thruster and power processor for New Millennium's Deep Space 1 mission. Technical Report NASA/TM-113129 (AIAA-97-2778), NASA, 1997. URL https://ntrs.nasa.gov/api/citations/19980017550/downloads/19980017550.pdf.

[5] Wiley J. Larson and James R. Wertz. *Space Mission Analysis and Design*. Microcosm Press / Kluwer, 3rd edition. Local copy; tables and pages listed in smad/sources.md were read.

[6] R. Shastry, D. A. Herman, G. C. Soulas, and M. J. Patterson. End-of-test performance and wear characterization of NEXT long-duration test. Technical report, NASA Glenn Research Center, 2014. URL https://ntrs.nasa.gov/api/citations/20140017457/downloads/20140017457.pdf.

[7] K. K. de Groh, B. A. Banks, and C. A. Karniotis. NSTAR extended life test discharge chamber flake analyses. Technical Report NASA/TM-2005-213195, NASA, 2005. URL https://ntrs.nasa.gov/api/citations/20050160232/downloads/20050160232.pdf.

[8] Mohamed S. El-Genk and Jean-Michel Tournier. Conceptual design of a 100-kwe space nuclear reactor power system with high-power AMTEC. Technical report, STAIF-2003, AIP Conference Proceedings 654, 2003. Local PDF; Table 1 and text read.

[9] JPL Solar System Dynamics. Planetary physical parameters; keplerian elements for approximate positions of the major planets. ssd.jpl.nasa.gov, 2025.

[10] A. Boylston, S. Greenhalge, and M. Duchek. Nuclear electric propulsion for Uranus missions parametric assessment. In *AIAA ASCEND 2025*, 2025. NTRS 20250006522. The full text (file name "NEP for Ice Giants") was read for the 20-kWe mass-equipment list, the 14.18 km/s schedule and the specific-mass curves.

[11] Steven R. Oleson et al. COMPASS final report: Nuclear electric propulsion (NEP)-chemical vehicle 1.2. Technical Report NASA/TM-20210017131, NASA, 2021. URL https://ntrs.nasa.gov/api/citations/20210017131/downloads/TM-20210017131_errata.pdf. Selected sections read.

[12] C. E. Garner, M. D. Rayman, et al. In-flight operation of the Dawn ion propulsion system through completion of the final orbit around dwarf planet Ceres, 2018. NTRS 20210008613; abstract only (about 51,250 h, 416 kg Xe, 11.5 km/s).

[13] Lee S. Mason. A power conversion concept for the Jupiter Icy Moons Orbiter. Technical Report NASA/TM-2003-212596 (AIAA-2003-6007), NASA, 2003. URL https://ntrs.nasa.gov/api/citations/20030105581/downloads/20030105581.pdf. Full text read (OCR noisy).

[14] Lee Mason, Steven Oleson, Paul Schmitz, Louis Qualls, Mitchell Smith, Brandon Ade, and Jose Navarro. Nuclear power concepts for high-power electric propulsion missions to Mars. In *Nuclear and Emerging Technologies for Space (NETS)*, 2021. URL https://ntrs.nasa.gov/api/citations/20210010243/downloads/NETS-2021-MASON%20final.pdf. Full text read.

[15] W. O. Miller and W. Shih. Lightweight carbon-carbon high-temperature space radiator. NASA Tech Briefs, September 2008 (NTRS 20080048181), 2008. URL https://ntrs.nasa.gov/citations/20080048181.

[16] G. Karpati, T. Hyde, H. Peabody, et al. Resource management and contingencies in aerospace concurrent engineering. Technical Report NTRS 20120013284, NASA Goddard Space Flight Center, 2012. URL https://ntrs.nasa.gov/api/citations/20120013284/downloads/20120013284.pdf. Text read; the margin-by-maturity table is an image and was not read.

[17] M. Chaiken. The Kilopower space nuclear fission power reactor. In *NASA Glenn Research Center*, 2019. URL https://ntrs.nasa.gov/api/citations/20190026448/downloads/20190026448.pdf.

[18] J. A. Hamley. Direct drive options for electric propulsion systems. Technical Report NASA TM 106576 (AIAA-95-364), NASA, 1995. URL https://ntrs.nasa.gov/api/citations/19950021440/downloads/19950021440.pdf.

[19] W. A. Hoskins, D. Homiak, R. J. Cassady, et al. Direct drive hall thruster system development. In *AIAA-2003-4726*, 2003. URL https://ntrs.nasa.gov/api/citations/20050207431/downloads/20050207431.pdf.

[20] J. D. Frieman, H. Kamhawi, et al. Completion of the long duration wear test of the NASA HERMeS hall thruster. In *AIAA Propulsion and Energy Forum*, 2019. URL https://ntrs.nasa.gov/api/citations/20190030504/downloads/20190030504.pdf.

[21] NASA Jet Propulsion Laboratory. NASA's optical comms demo transmits data over 140 million miles, 2024. URL https://www.jpl.nasa.gov/news/nasas-optical-comms-demo-transmits-data-over-140-million-miles/. Accessed 2026-11-04.

[22] NASA Jet Propulsion Laboratory. Deep space optical communications (DSOC), . URL https://www.jpl.nasa.gov/missions/dsoc. Accessed 2026-11-04.

[23] Amy A. Simon et al. Uranus orbiter and probe: Planetary mission concept study for the 2023–2032 decadal survey. Technical report, NASA Goddard Space Flight Center and Johns Hopkins Applied Physics Laboratory, 2021. URL https://science.nasa.gov/wp-content/uploads/2023/10/uranus-orbiter-and-probe.pdf. Accessed 2026-11-04.

[24] NASA Office of Inspector General. Audit of NASA's deep space network. Technical Report IG-23-016, NASA OIG, 2023. URL https://oig.nasa.gov/docs/IG-23-016.pdf. Accessed 2026-11-04.

[25] NASA. NASA says farewell to MAVEN mars mission, hosts media call today, 2026. URL https://www.nasa.gov/news-release/nasa-says-farewell-to-maven-mars-mission-hosts-media-call-today/. Accessed 2026-11-04.

[26] NASA. NASA's new horizons spacecraft begins intensive data downlink phase, 2015. URL https://www.nasa.gov/general/nasas-new-horizons-spacecraft-begins-intensive-data-downlink-phase/. Accessed 2026-11-04.

[27] American Nuclear Society. NASA announces plan for space nuclear propulsion by 2028, March 2026. URL https://www.ans.org/news/2026-03-25/article-7879/nasa-announces-plan-for-space-nuclear-propulsion-by-2028/. Nuclear Newswire. Accessed 2026-11-04.

[28] Robert E. Thomas, Rittu S. Raju, and James J. Bontempo. Characterizing electrical arc behavior in ion engines. In *AIAA SciTech Forum*, 2025. NASA NTRS 20240014086. https://ntrs.nasa.gov/citations/20240014086. Accessed 2026-11-04.

[29] U.S. Department of Energy. SNAP-10A (energy.gov feature). URL https://www.energy.gov/node/4852675. Accessed 2026-11-04.

[30] Jeff Foust. NASA and DARPA select Lockheed Martin to develop DRACO nuclear propulsion demo. SpaceNews, July 2023. URL https://spacenews.com/nasa-and-darpa-select-lockheed-martin-to-develop-draco-nuclear-propulsion-demo/. Accessed 2026-11-04.

[31] The Planetary Society. Europa clipper cost chart. URL https://planetary.org/charts/europa-clipper-cost-chart. Accessed 2026-11-04.

[32] Office of Management and Budget. President trump's fiscal year 2026 discretionary funding request (NASA excerpts), May 2025. URL https://www.nasa.gov/wp-content/uploads/2025/05/fiscal-year-2026-discretionary-budget-request-nasa-excerpts.pdf. Accessed 2026-11-04.

[33] NASA Science. NASA's dragonfly rotorcraft mission to saturn's moon titan confirmed, 2024. URL https://science.nasa.gov/missions/dragonfly/nasas-dragonfly-rotorcraft-mission-to-saturns-moon-titan-confirmed/. Accessed 2026-11-04.

[34] NASA Office of Inspector General. NASA's management of the human landing system contracts. Technical Report IG-26-004, NASA OIG, March 2026. URL https://oig.nasa.gov/wp-content/uploads/2026/03/final-report-ig-26-004-nasas-management-of-the-human-landing-system-contracts.pdf. Accessed 2026-11-04.

[35] NASA. Space reactor-1 freedom (mission page), 2026. URL https://www.nasa.gov/mission/space-reactor-1-freedom/. Last updated 2026-11-03. Accessed 2026-11-04.

[36] NASA/JPL. Dawn launch: Mission to vesta and ceres (press kit), September 2007. URL https://www.jpl.nasa.gov/news/press_kits/dawn-launch.pdf. Accessed 2026-11-04.

[37] Marc D. Rayman and Charles E. Garner. In-flight operation of the dawn ion propulsion system through completion of the final orbit around dwarf planet ceres. NASA NTRS 20210008613, JPL preprint, 2018. URL https://ntrs.nasa.gov/citations/20210008613. Accessed 2026-11-04.

[38] NASA/JPL. Psyche press kit: Spacecraft. URL https://www.jpl.nasa.gov/press-kits/psyche/mission/spacecraft/. Accessed 2026-11-04.

[39] NASA/JPL. NASA's psyche mission aces mars flyby, targets metal-rich asteroid, 2026. URL https://www.jpl.nasa.gov/news/nasas-psyche-mission-aces-mars-flyby-targets-metal-rich-asteroid/. Accessed 2026-11-04.

[40] ESA. Electric blue thrusters propelling bepicolombo to mercury, 2018. URL https://www.esa.int/Enabling_Support/Space_Engineering_Technology/Electric_blue_thrusters_propelling_BepiColombo_to_Mercury. Accessed 2026-11-04.

[41] ESA. Bepicolombo begins mercury arrival with MTM separation success, September 2026. URL https://www.esa.int/Science_Exploration/Space_Science/BepiColombo/BepiColombo_begins_Mercury_arrival_with_MTM_separation_success. Accessed 2026-11-04.

[42] Joe Cassady, Brian Boyce, Brent Kearney, Dan Herman, and Peter Peterson. Development of 12 kw hall thrusters for NASA lunar gateway power and propulsion element. In *8th Space Propulsion Conference*, Estoril, Portugal, 2022. NASA NTRS 20220007037. https://ntrs.nasa.gov/citations/20220007037. Accessed 2026-11-04.

[43] NASA. NASA unveils initiatives to achieve america's national space policy, March 2026. URL https://www.nasa.gov/news-release/nasa-unveils-initiatives-to-achieve-americas-national-space-policy/. Accessed 2026-11-04.

[44] NASA. NASA utilization of space nuclear systems for robotic and human exploration missions: Response to EO 13972. Technical report, NASA, July 2022. URL https://www.nasa.gov/wp-content/uploads/2023/04/nasa-utilization-of-space-nuclear-systems.pdf. Accessed 2026-11-04.

[45] David Buden. Summary of space nuclear reactor power systems (1983–1992). Technical report, Idaho National Engineering Laboratory, 1993. URL https://www.osti.gov/servlets/purl/10151265. Accessed 2026-11-04.

[46] Lee S. Mason. A power conversion concept for the Jupiter Icy Moons Orbiter. In *1st International Energy Conversion Engineering Conference*, number AIAA-2003-6007, Portsmouth, VA, 2003. NASA NTRS 20150012286. https://ntrs.nasa.gov/api/citations/20150012286/downloads/20150012286.pdf. Accessed 2026-11-04.

[47] U.S. Government Accountability Office. NASA's space vision: Business case for prometheus 1 needed to ensure requirements match available resources. Technical Report GAO-05-242, GAO, 2005. URL https://www.gao.gov/products/gao-05-242. Accessed 2026-11-04.

[48] David I. Poston, Marc A. Gibson, Rene G. Sanchez, and Patrick R. McClure. Results of the KRUSTY nuclear system test. *Nuclear Technology*, 206(sup1):89–117, 2020. doi: 10.1080/00295450.2020.1730673. Open copy: https://www.osti.gov/pages/servlets/purl/1648091. Accessed 2026-11-04.

[49] NASA. NASA announces artemis concept awards for nuclear power on moon (release 22-062), June 2022. URL https://www.nasa.gov/?p=129588. Accessed 2026-11-04.

[50] Nuclear Engineering International. NASA fast-tracks lunar reactor plan, August 2025. URL https://www.neimagazine.com/news/nasa-fast-tracks-lunar-reactor-plan/. Accessed 2026-11-04.

[51] New Space Economy. What does NASA's SPARC lunar reactor procurement reveal about lunar reactor-1?, September 2026. URL https://newspaceeconomy.ca/2026/09/01/what-does-nasas-sparc-lunar-reactor-procurement-reveal-about-lunar-reactor-1/. Secondary analysis of SAM.gov draft solicitation (sam.gov/opp/e11e300dbee845d195c583ff87afa81a). Accessed 2026-11-04.

[52] Breaking Defense. DARPA's DRACO nuclear propulsion project roars no more, June 2025. URL https://breakingdefense.com/2025/06/darpas-draco-nuclear-propulsion-project-roars-no-more/. Accessed 2026-11-04.

[53] Anatoly Zak. Nuclear-powered space tug (TEM/zevs). RussianSpaceWeb. URL https://www.russianspaceweb.com/tem.html. Last updated 2024-07-11. Accessed 2026-11-04.

[54] NASA Jet Propulsion Laboratory. Mars 2020 perseverance landing press kit: Mission, . URL https://www.jpl.nasa.gov/news/press_kits/mars_2020/landing/mission/. Accessed 2026-11-04.

[55] Tom Percy, Melissa McGuire, and Tara Polsgrove. Combining solar electric propulsion and chemical propulsion for crewed missions to Mars. In *IEEE Aerospace Conference (NTRS 20150006952)*, 2015. URL https://ntrs.nasa.gov/api/citations/20150006952/downloads/20150006952.pdf. Accessed 2026-11-04.

[56] Carolyn R. Mercer, Melissa L. McGuire, Steven R. Oleson, and Michael J. Barrett. Solar electric propulsion concepts for human space exploration. In *AIAA SPACE 2015 (NTRS 20150019856)*, 2015. URL https://ntrs.nasa.gov/api/citations/20150019856/downloads/20150019856.pdf. Accessed 2026-11-04.

[57] Spaceflight Now. Spacex launches $5.2 billion nasa mission to jupiter's ocean moon, europa, 2024. URL https://spaceflightnow.com/2024/10/14/live-coverage-spacex-to-launch-nasas-europa-clipper-on-falcon-heavy-rocket-from-the-kenney-space Accessed 2026-11-04.

[58] NASA Science. Europa clipper: Mission overview. URL https://science.nasa.gov/mission/europa-clipper/mission-overview/. Accessed 2026-11-04.

[59] NASA/JPL. Europa clipper press kit: Spacecraft, 2024. URL https://www.jpl.nasa.gov/press-kits/europa-clipper/mission/spacecraft/. Accessed 2026-11-04.

[60] NASA Jet Propulsion Laboratory. Juno launch press kit: Quick facts, . URL https://www.jpl.nasa.gov/news/press_kits/juno/facts/. Accessed 2026-11-04.

[61] European Space Agency. Juice factsheet. URL https://www.esa.int/Science_Exploration/Space_Science/Juice/Juice_factsheet. Accessed 2026-11-04.

[62] deutschland.de. Groundbreaking esa mission to jupiter. URL https://deutschland.de/en/topic/knowledge/jupiter-mission-juice-european-space-agency-esa-germany-dlr. Accessed 2026-11-04.

[63] Spaceflight Now. Robotic mission to uranus receives top priority in decadal survey, 2022. URL https://spaceflightnow.com/2022/04/19/robotic-mission-to-uranus-receives-top-priority-in-decadal-survey/. Accessed 2026-11-04.

[64] Amy A. Simon, Richard C. Anderson, Daniel T. Gallagher, Michael J. Amato, Donald H. Ellison, Jacob A. Englander, et al. Uranus orbiter and probe: Mission challenges and concept updates since the origins, worlds, and life decadal survey. *Planetary Science Journal (submitted Dec. 2025, revised Mar. 2026; NTRS 20260004996)*, 2026. URL https://ntrs.nasa.gov/api/citations/20260004996/downloads/SimonUranusSTI.pdf. Accessed 2026-11-04.

[65] Soumyo Dutta, Warren Davis, Rohan G. Deshmukh, Andrew Gomez-Delrio, Rafael A. Lugo, James B. Scoggins, Eli R. Shellabarger, et al. Aerocapture solutions for flagship-class uranus orbiter and probe (ntrs 20240014560), 2024. URL https://ntrs.nasa.gov/api/citations/20240014560/downloads/Aerocapture_Overview_v3.pdf. Accessed 2026-11-04.

[66] Marcia Smith. Musk embraces the moon. SpacePolicyOnline, February 2026. URL https://spacepolicyonline.com/news/musk-embraces-the-moon/. Accessed 2026-11-04.

[67] K. A. White III. Liquid droplet radiator development status. Technical Report NASA TM 89852, NASA, 1987. URL https://ntrs.nasa.gov/api/citations/19870010920/downloads/19870010920.pdf.

[68] L. Soffer and G. N. Wright. Effect of reactor coolant radioactivity upon configuration feasibility for a nuclear electric propulsion vehicle. Technical Report NASA TM X-2949, NASA Lewis, 1973. URL https://ntrs.nasa.gov/api/citations/19740004251/downloads/19740004251.pdf.

[69] A. T. Mattick and A. Hertzberg. The liquid droplet radiator: An ultralightweight heat rejection system for efficient energy conversion in space. In *IAF-81-185*, 1981. NTRS 19810062992; abstract only.

[70] S. R. Oleson. An analytical optimization of electric propulsion orbit transfer vehicles. Technical Report NASA CR-191129, NASA, 1993. NTRS 19930017871; abstract only.

[71] N. Berend. Interest of staged architectures with drop tanks for space vehicles using electric propulsion. In *International Astronautical Congress*, number IAC-15,C4,4, 2015. Abstract page only (ONERA); reports about 10–13% lower required specific mass at payload ratio 0.1.

[72] A. Simon, R. Anderson, et al. Uranus orbiter and probe: Mission challenges and concept updates since the origins, worlds, and life decadal survey. *Planetary Science Journal (submitted)*, 2026. URL https://ntrs.nasa.gov/api/citations/20260004996/downloads/SimonUranusSTI.pdf.

[73] J. Englander, D. Ellison, N. Hatten, K. Hughes, A. Mudek, and T. Rashied. Robust access to Uranus via solar electric propulsion. In *AAS*, 2025. NTRS 20250004387.

[74] M. Duchek, A. Boylston, S. Greenhalge, E. Grella, C. Harnack, W. Machemer, and A. Shaffer. NEP capabilities for Uranus and the outer planets: Power, maneuverability and faster transit times. Accelerating Space Science with Nuclear Technologies workshop, NTRS 20260003605, 2026. URL https://ntrs.nasa.gov/api/citations/20260003605/downloads/Uranus_Mission_ASSwNT_workshop_2026-05-07-AMA.pdf. Slide text read; row labels of the fastest-time table scrambled by text extraction.

[75] Jeffrey A. George. Multi-reactor power system configurations for multimegawatt nuclear electric propulsion. Technical Report NASA TM 105212, NASA, 1991. URL https://ntrs.nasa.gov/api/citations/19920000839/downloads/19920000839.pdf.